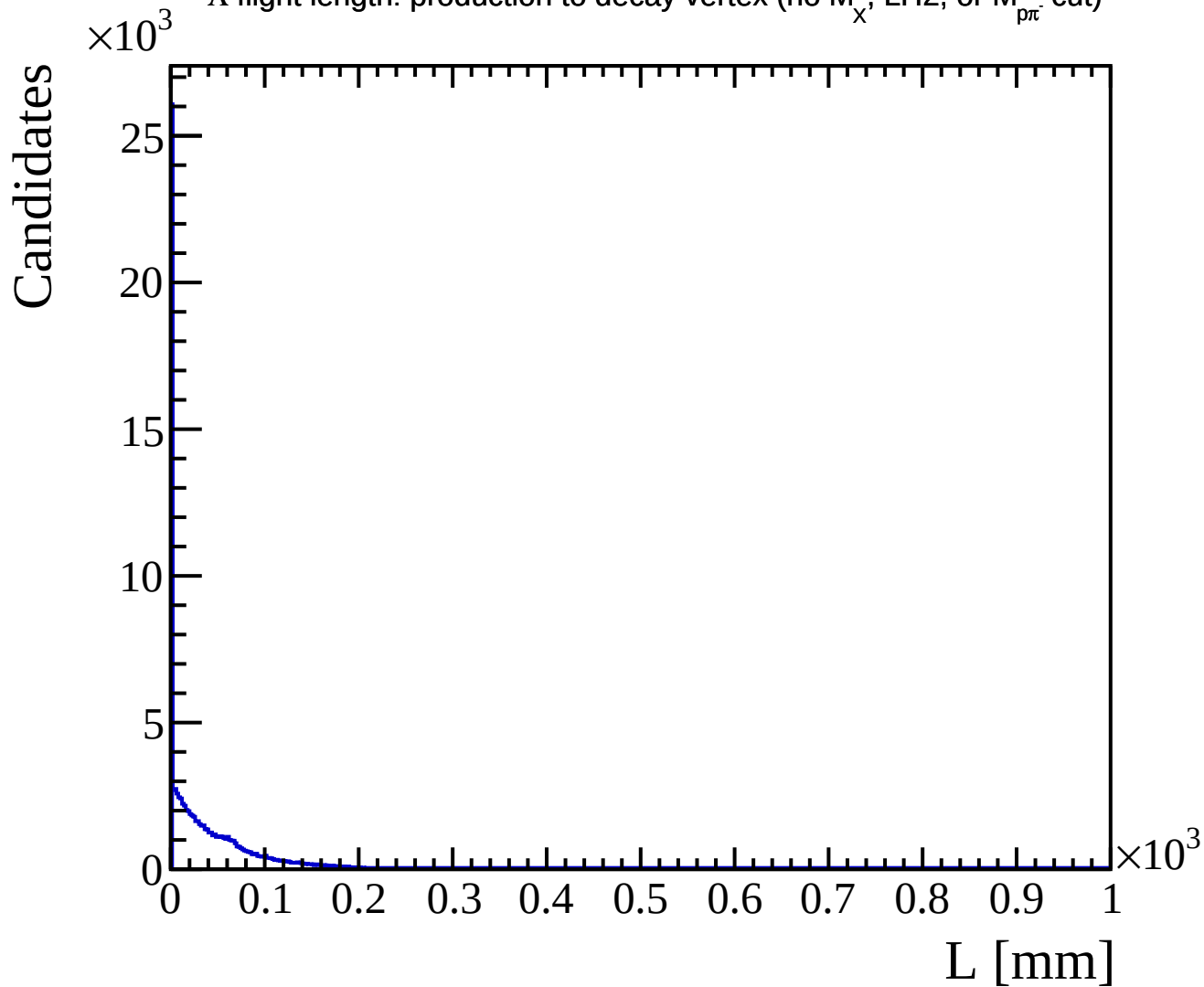
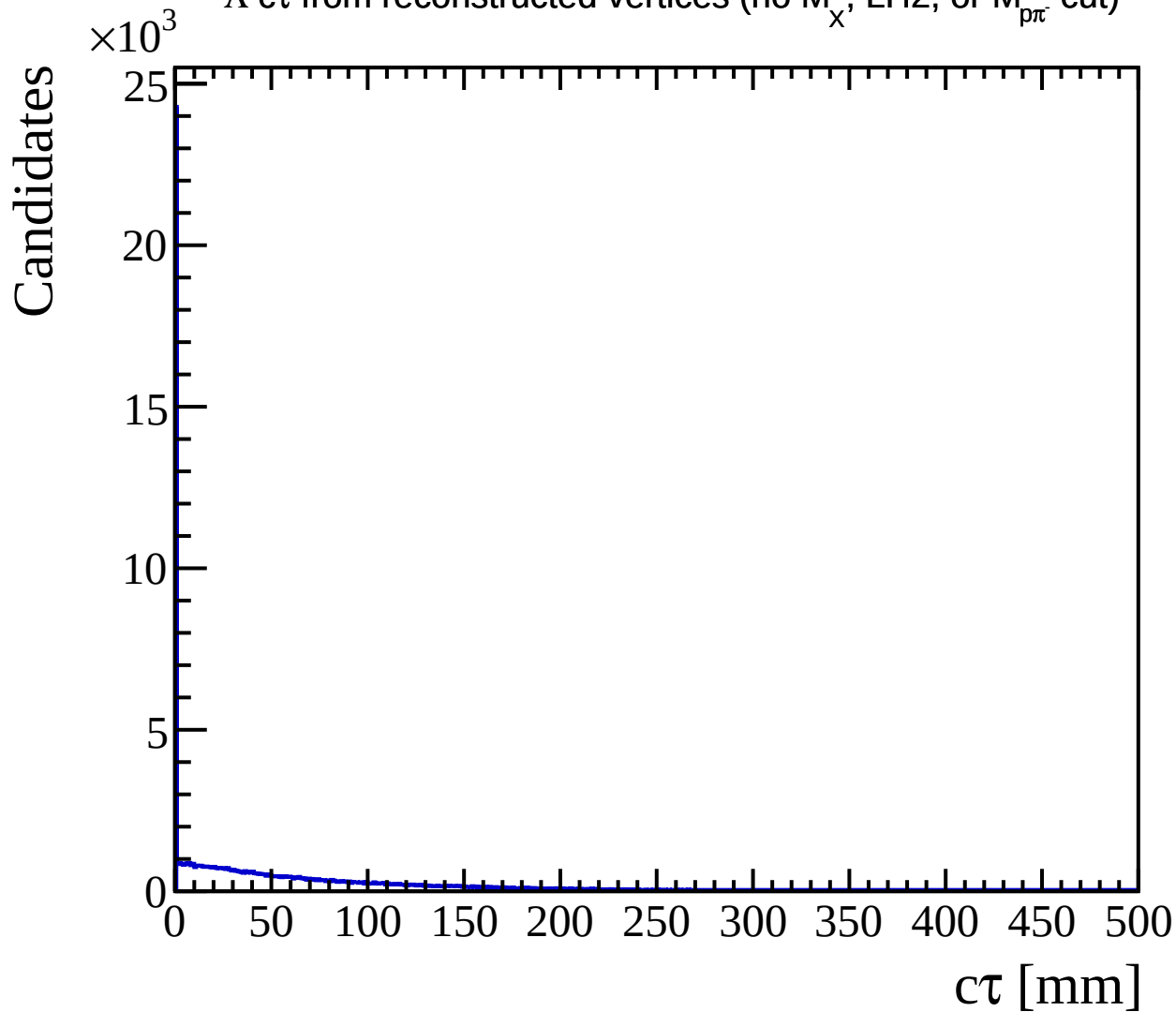


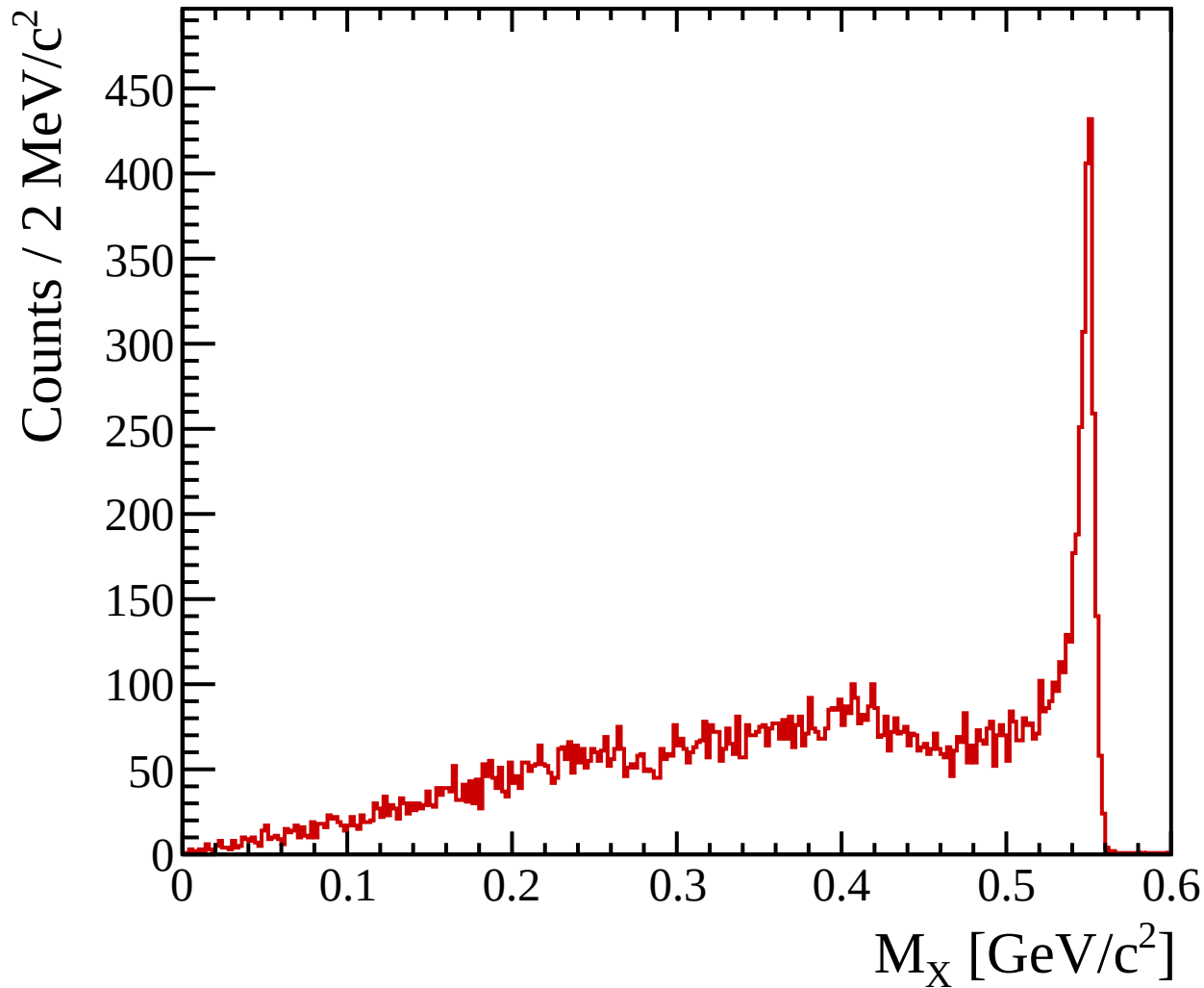
$\Lambda$  flight length: production to decay vertex (no  $M_X$ , LH2, or  $M_{p\pi^-}$  cut)



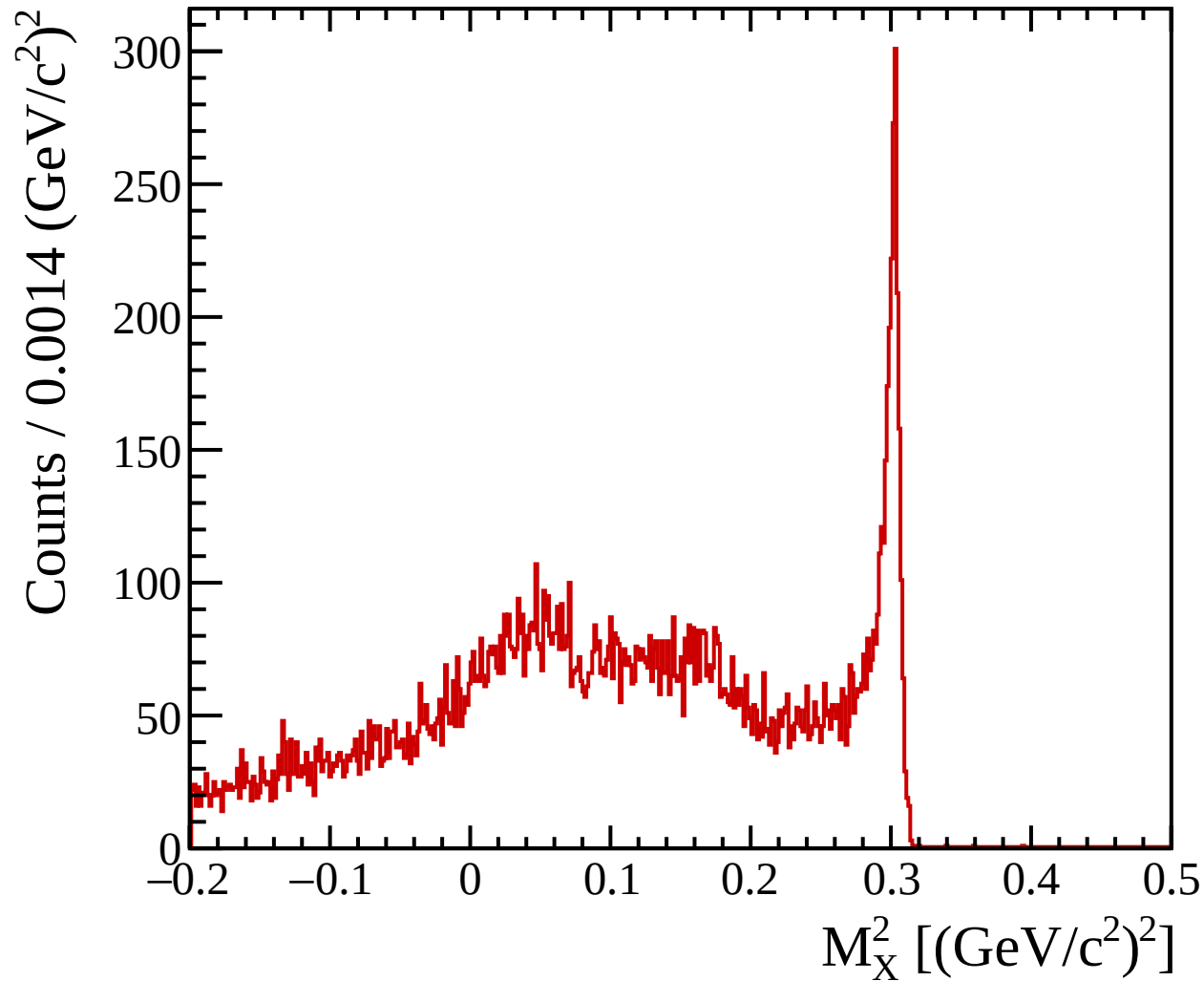
$\Lambda$   $c\tau$  from reconstructed vertices (no  $M_{\chi}$ , LH2, or  $M_{\rho\pi}$  cut)

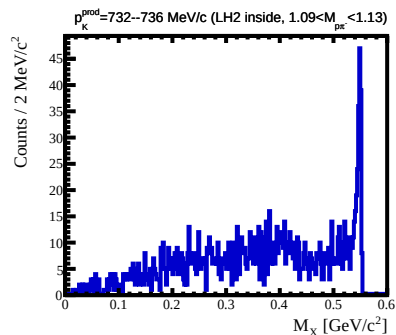
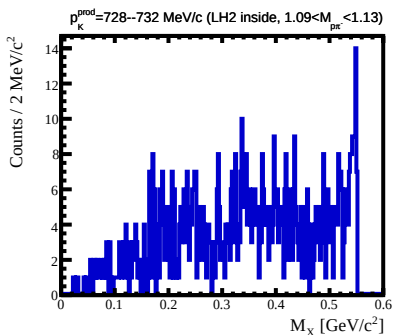
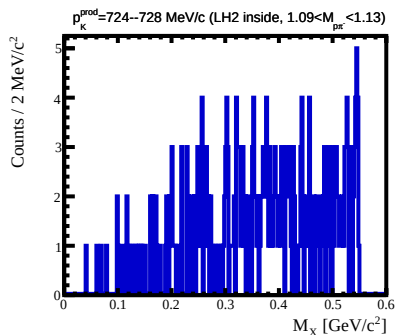
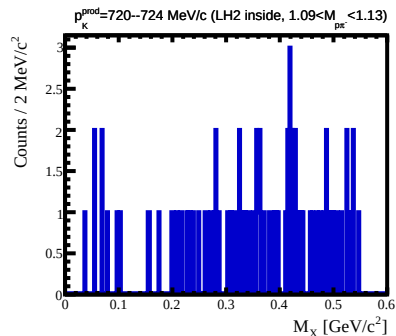
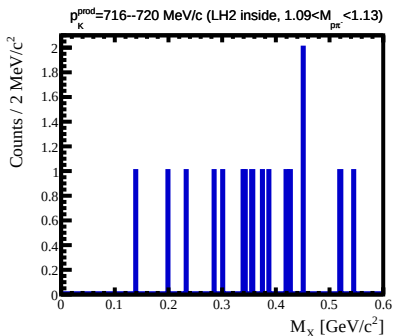
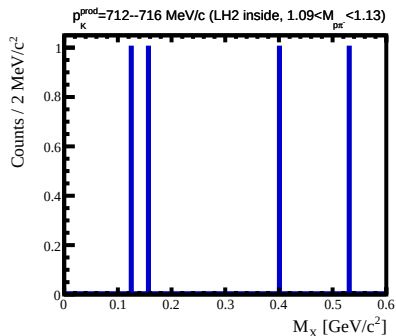
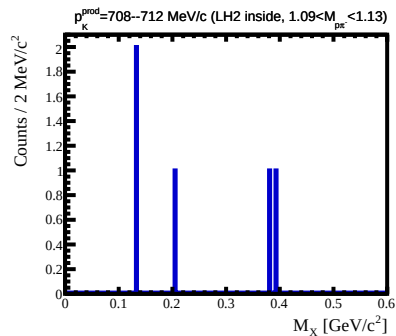
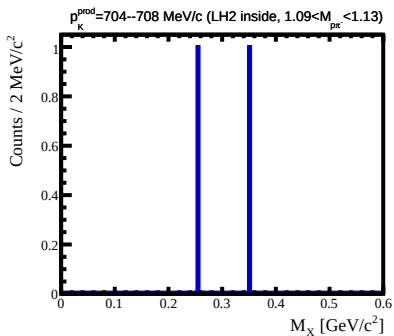
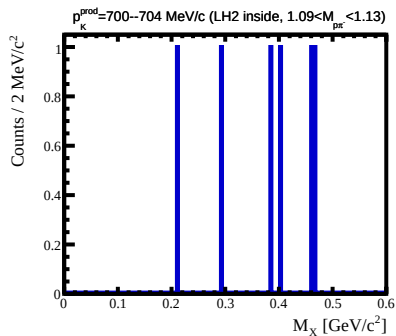


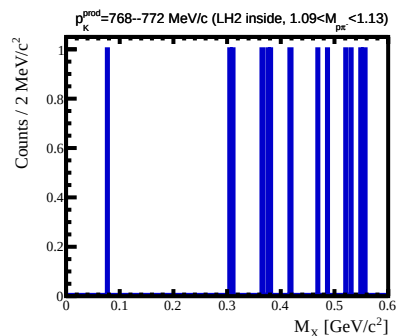
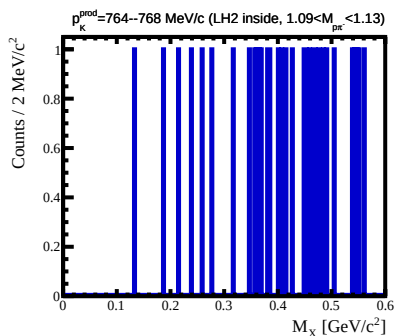
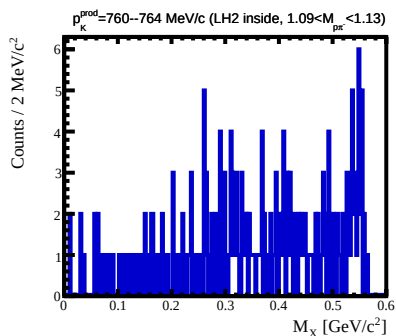
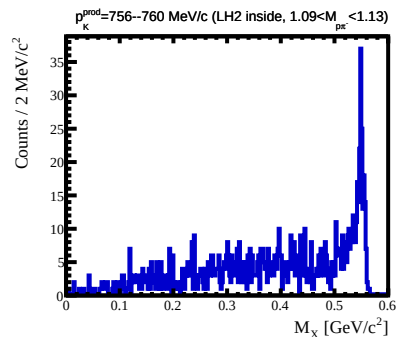
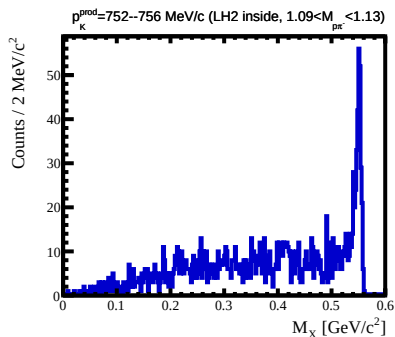
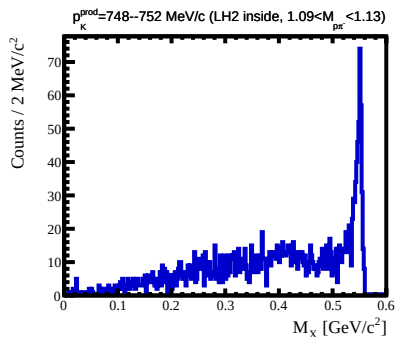
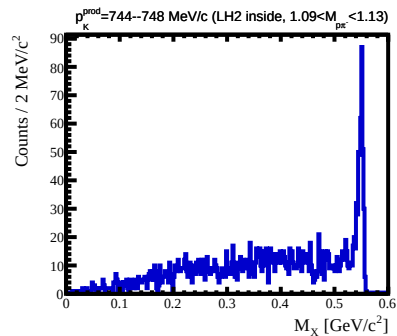
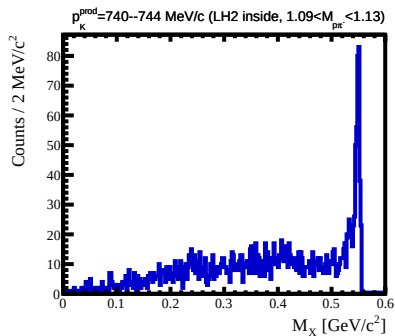
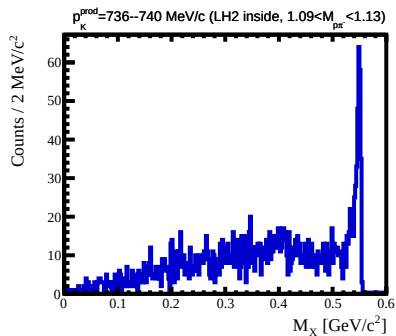
$\Lambda$  missing mass: LH2 inside,  $1.09 < M_{p\pi^-} < 1.13$

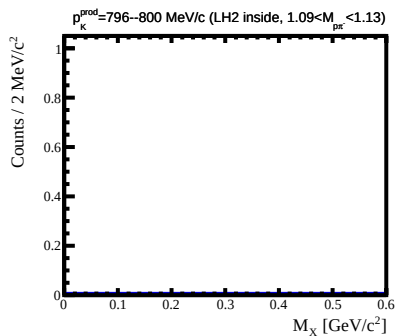
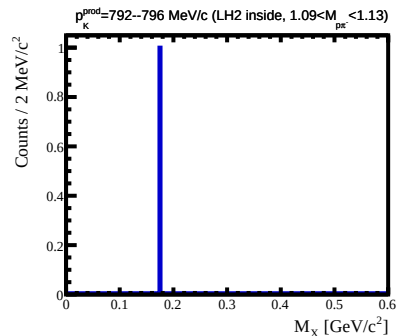
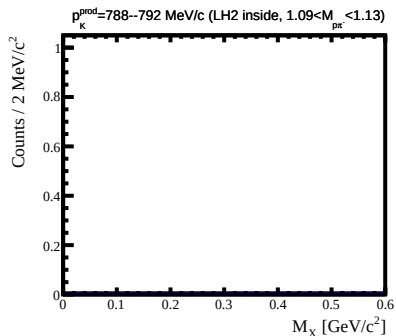
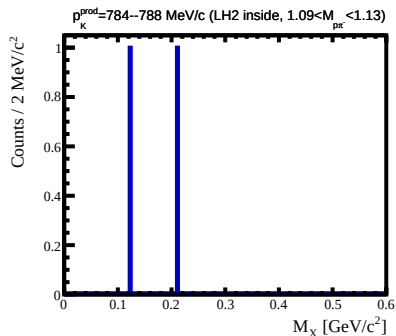
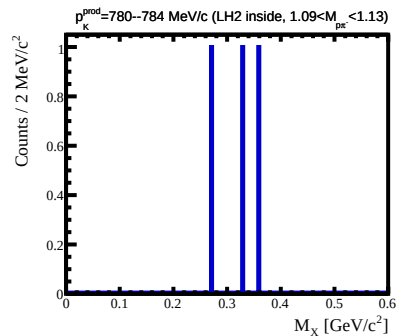
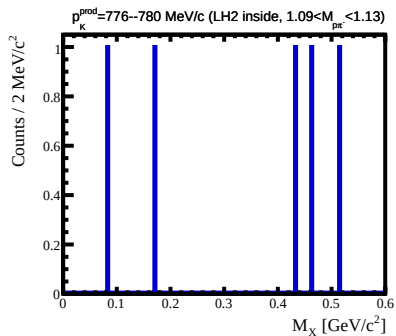
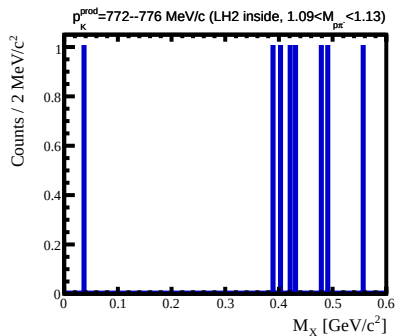


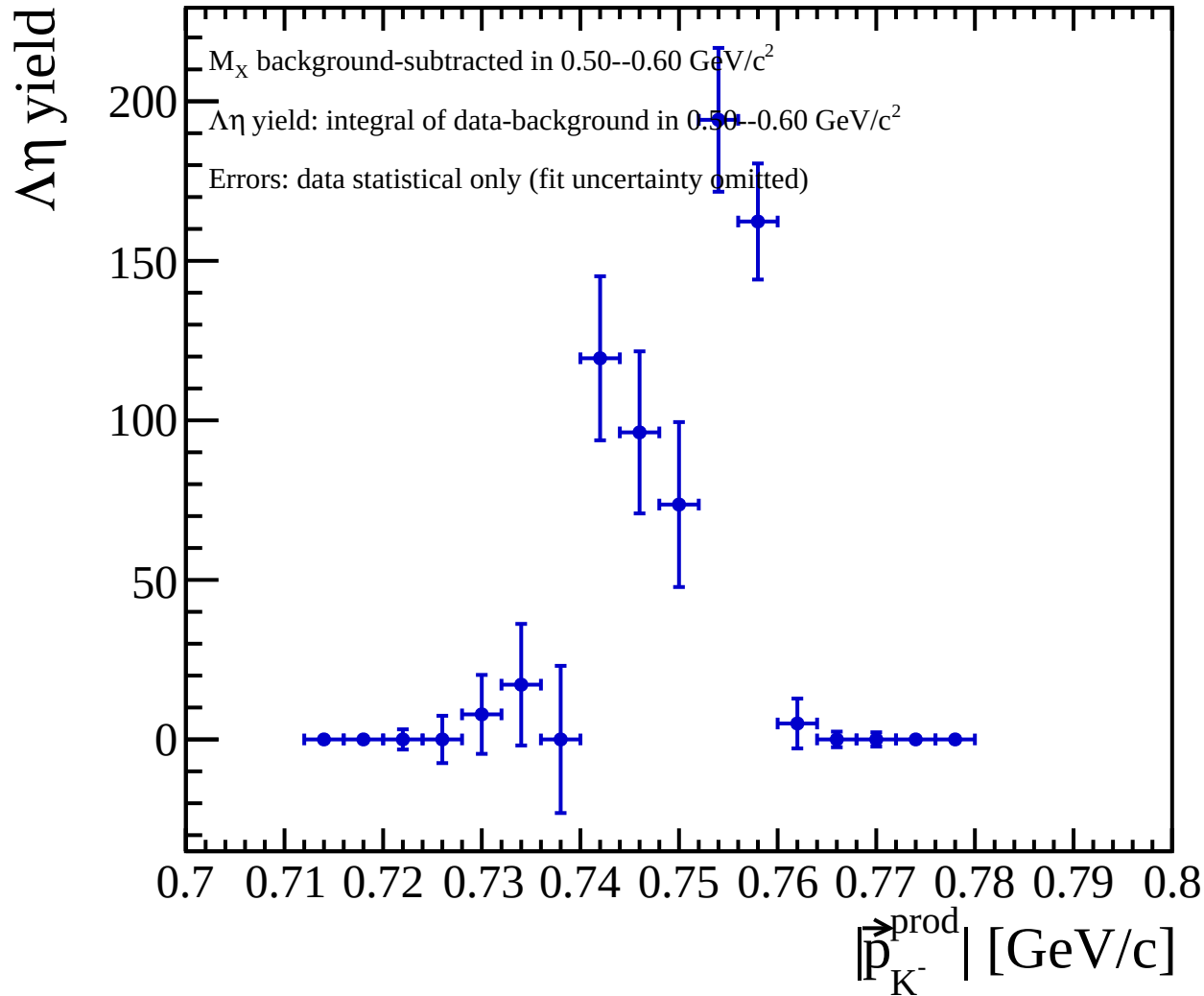
$\Lambda$  missing mass squared: LH2 inside,  $1.09 < M_{p\pi^-} < 1.13$



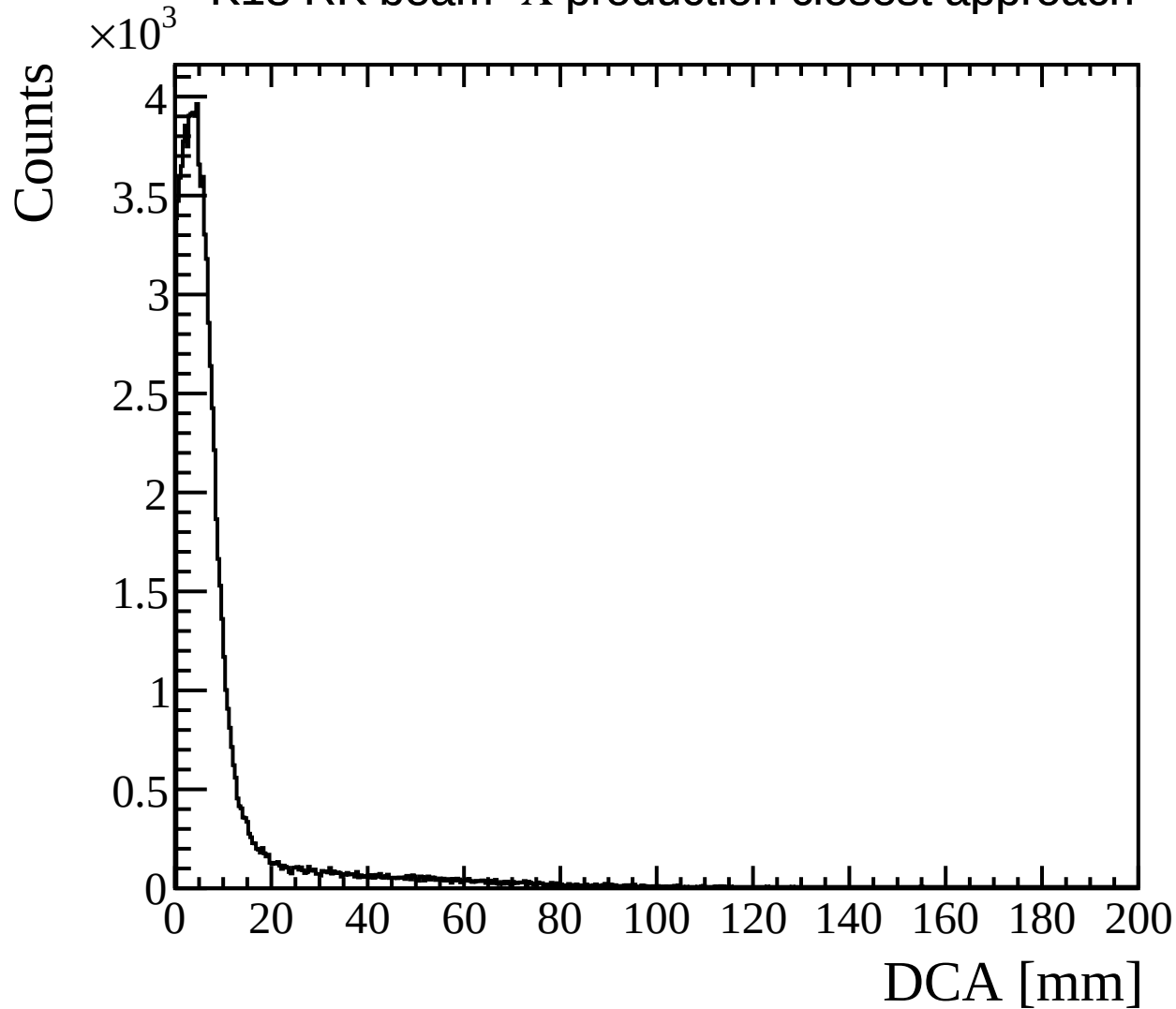




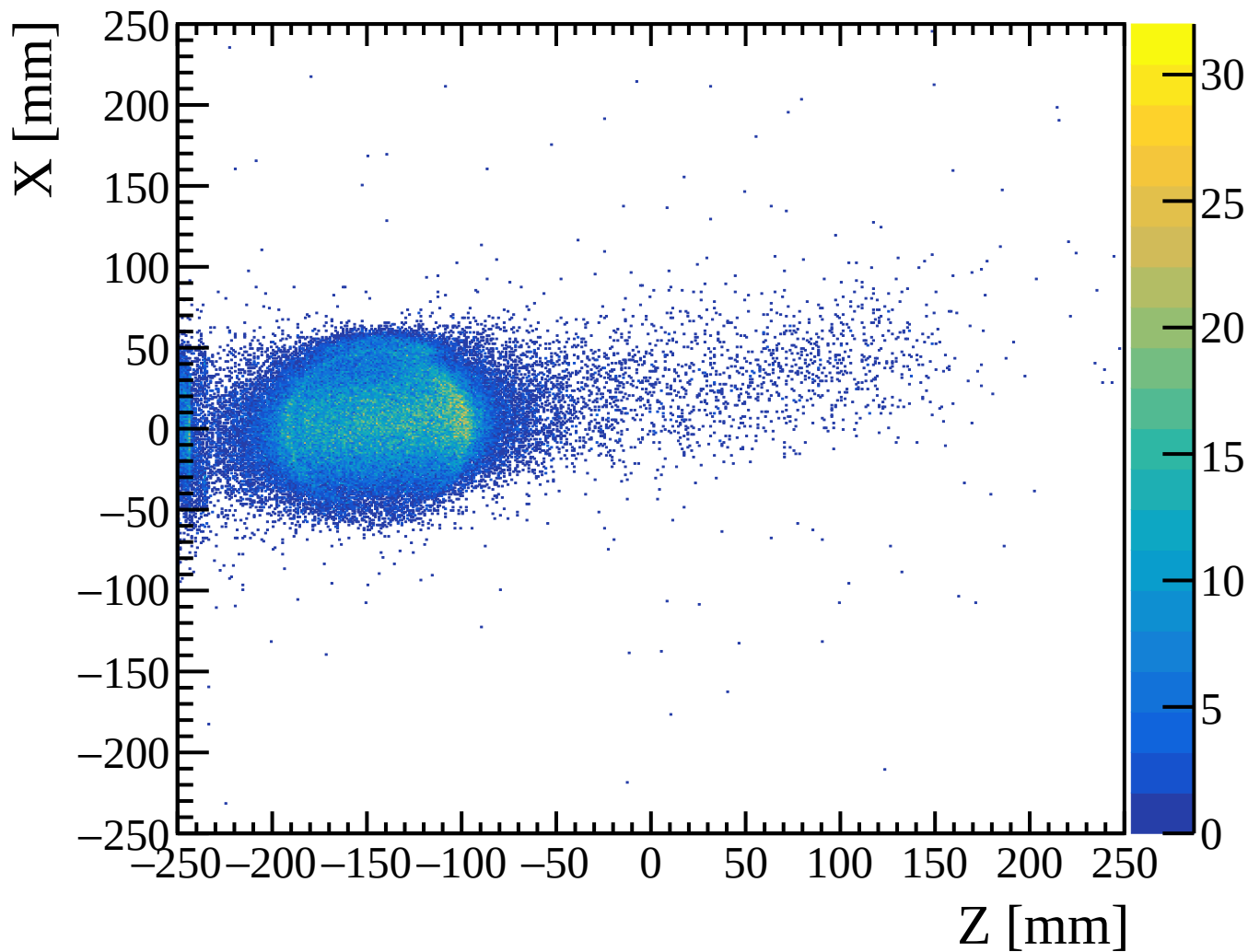




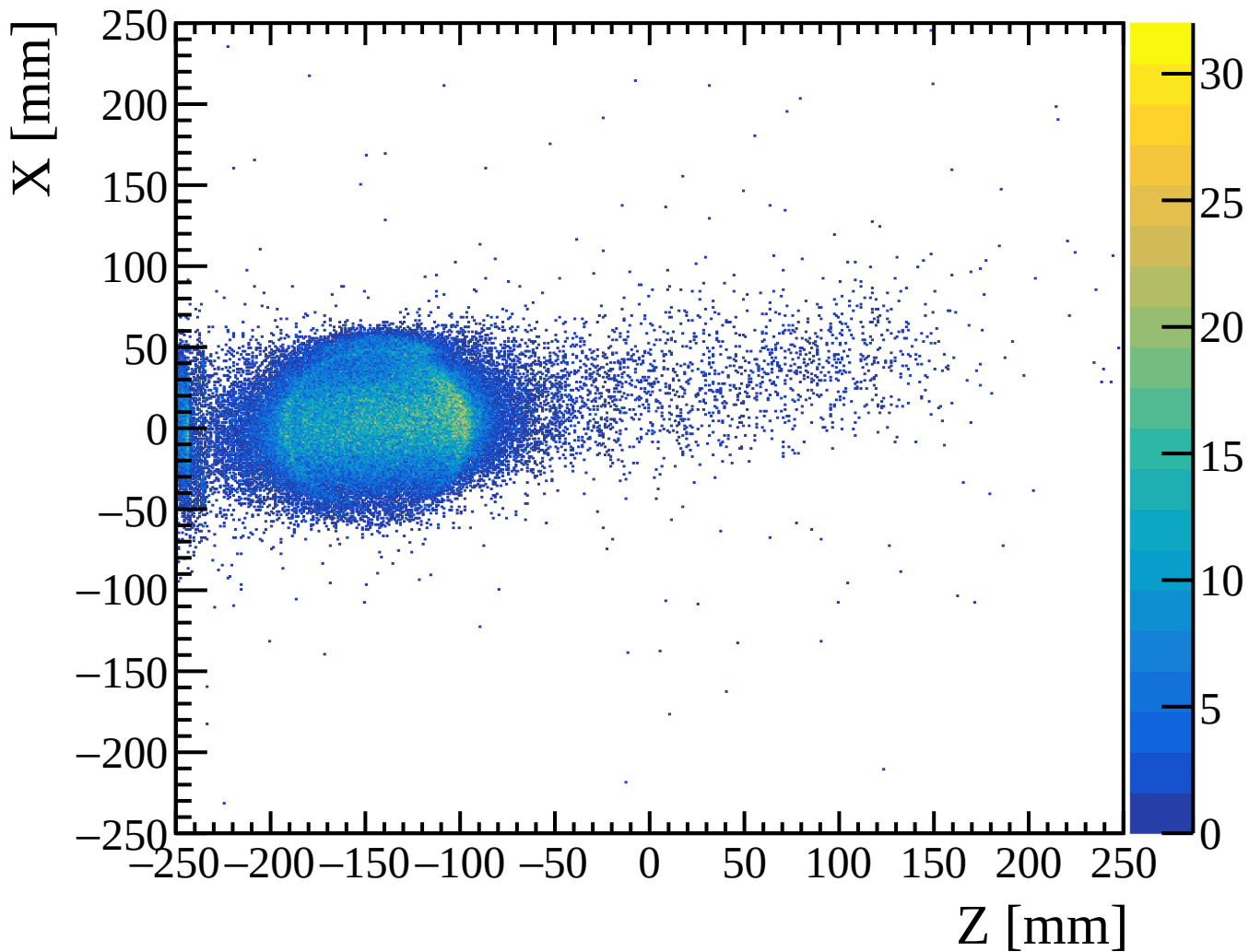
# K18 RK beam-- $\Lambda$ production closest approach



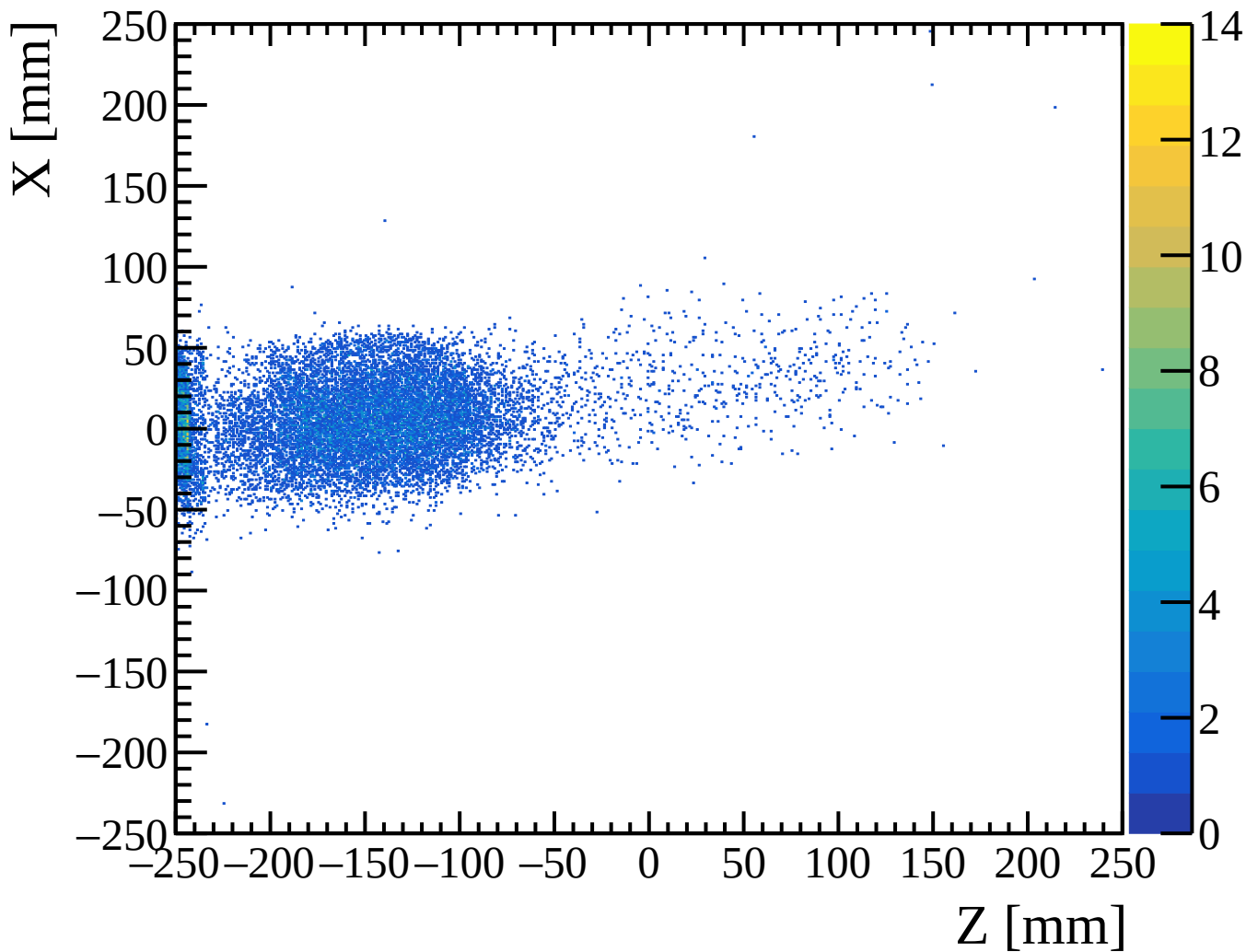
Reconstructed  $\Lambda$  production vertex (DST inclusive)



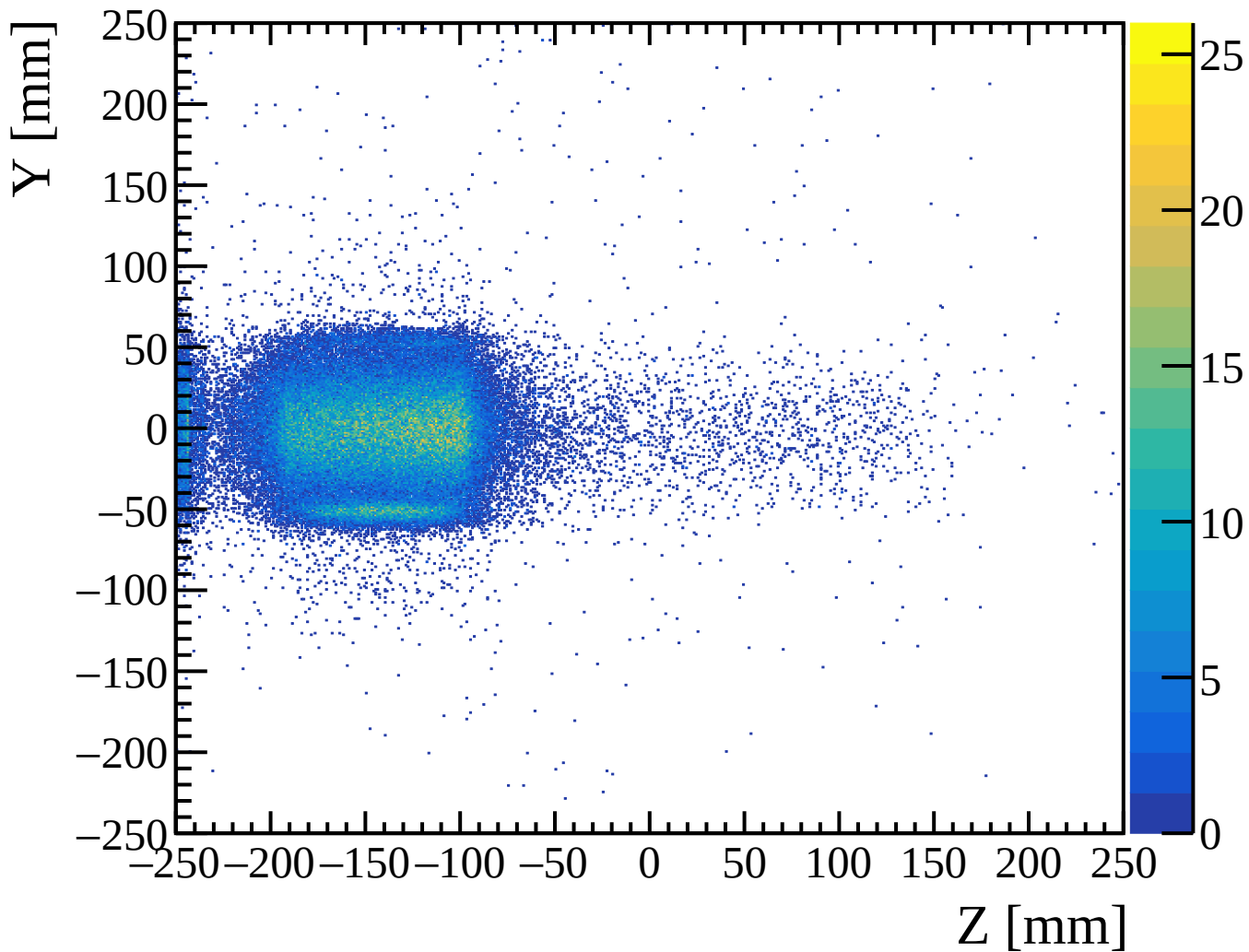
Reconstructed  $\Lambda$  production vertex (no LH2/Lambda-mass/ $M_x$  cut)



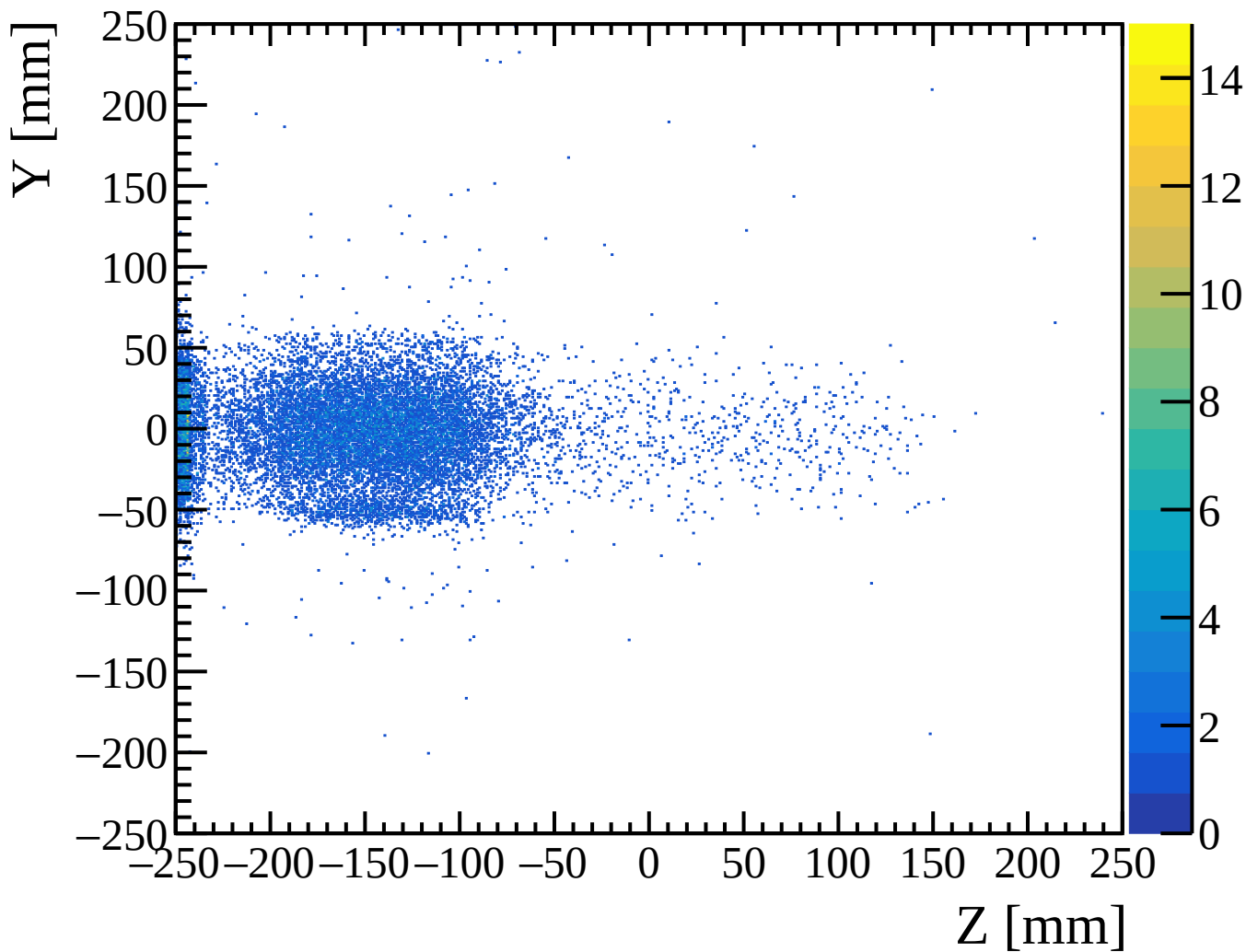
Reconstructed  $\Lambda$  production vertex (no LH2/Lambda-mass cut,  $M_x > 0.50$ )



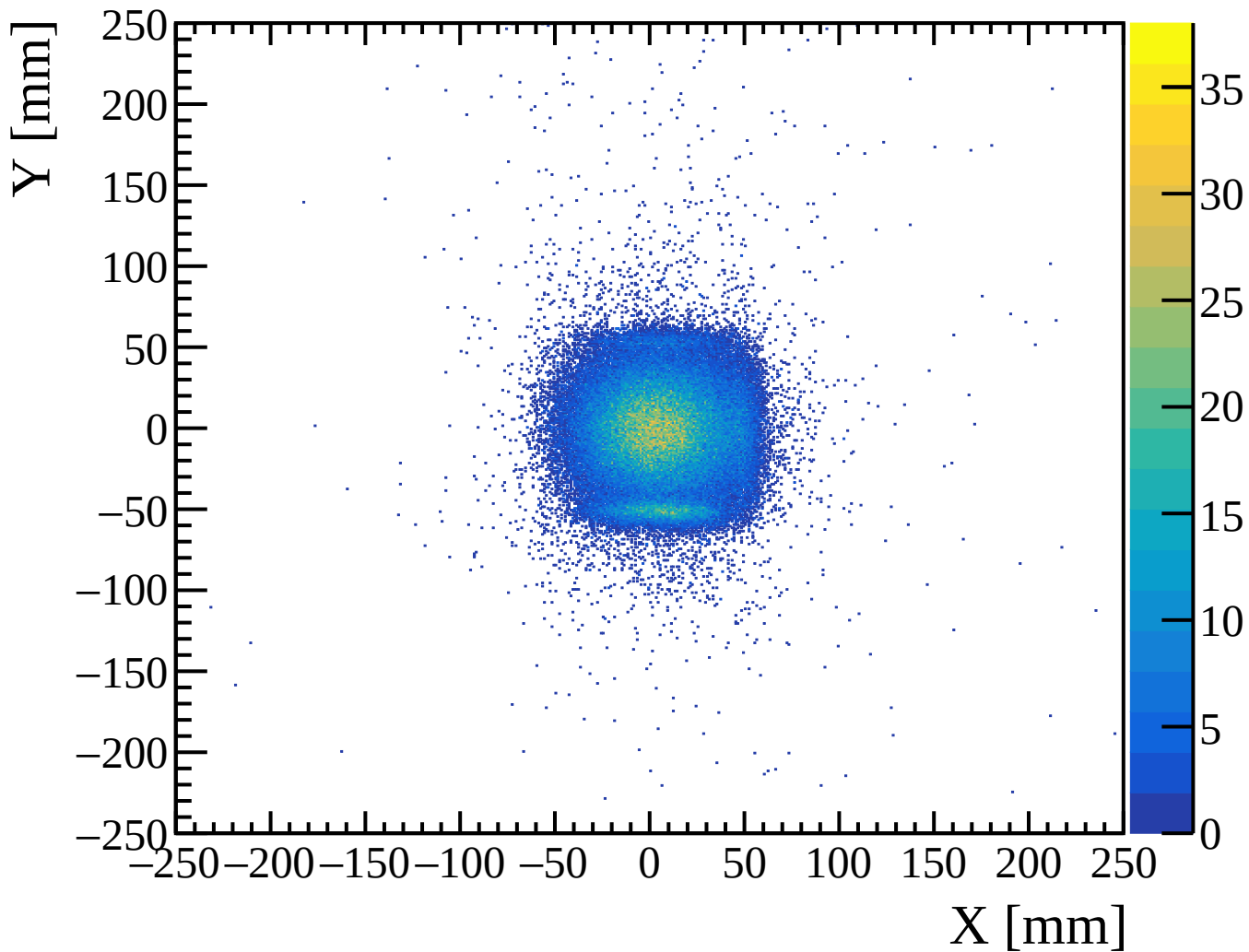
Reconstructed  $\Lambda$  production vertex (no LH2/Lambda-mass/ $M_x$  cut)



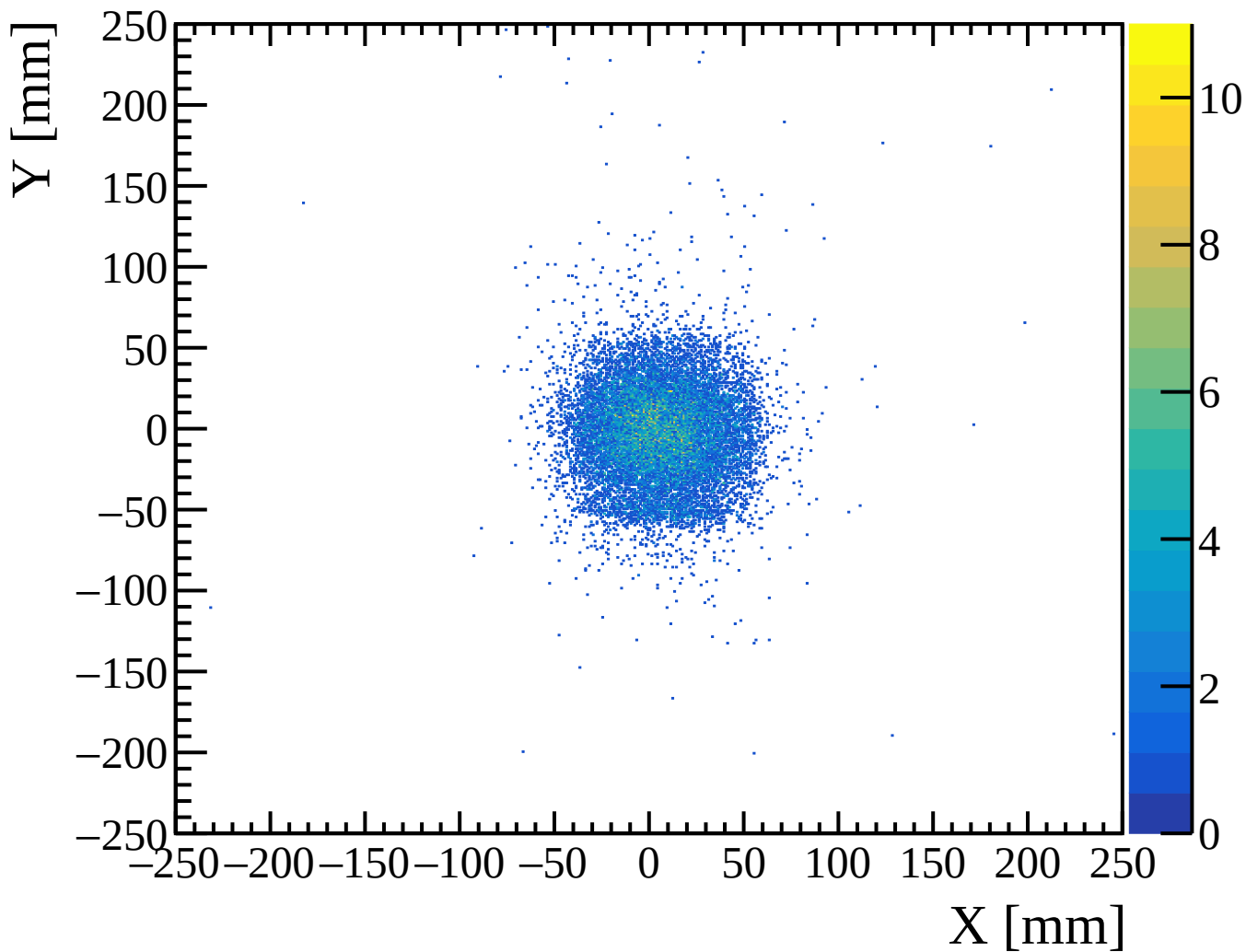
Reconstructed  $\Lambda$  production vertex (no LH2/Lambda-mass cut,  $M_x > 0.50$ )

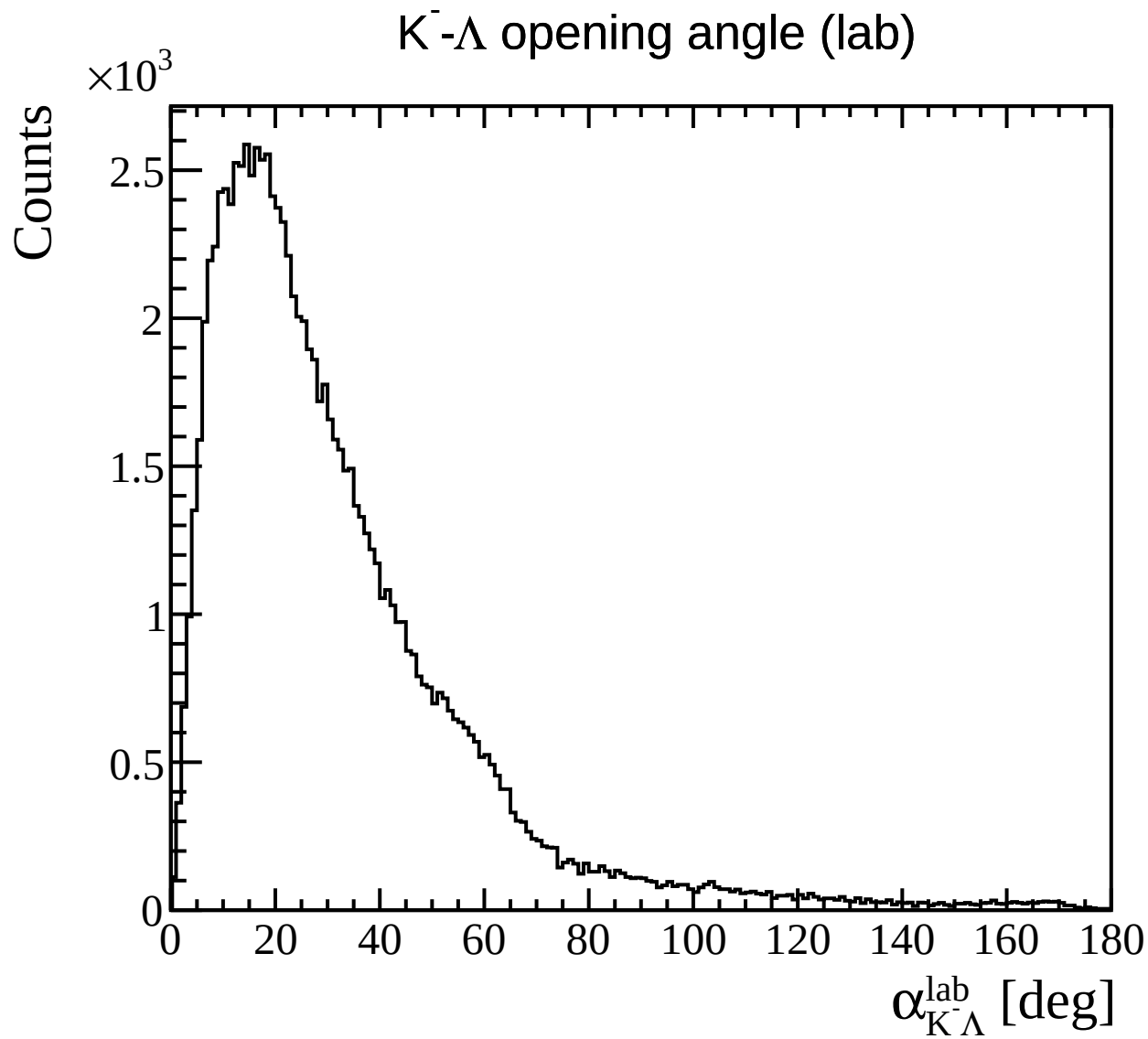


Reconstructed  $\Lambda$  production vertex (no LH2/Lambda-mass/ $M_x$  cut)

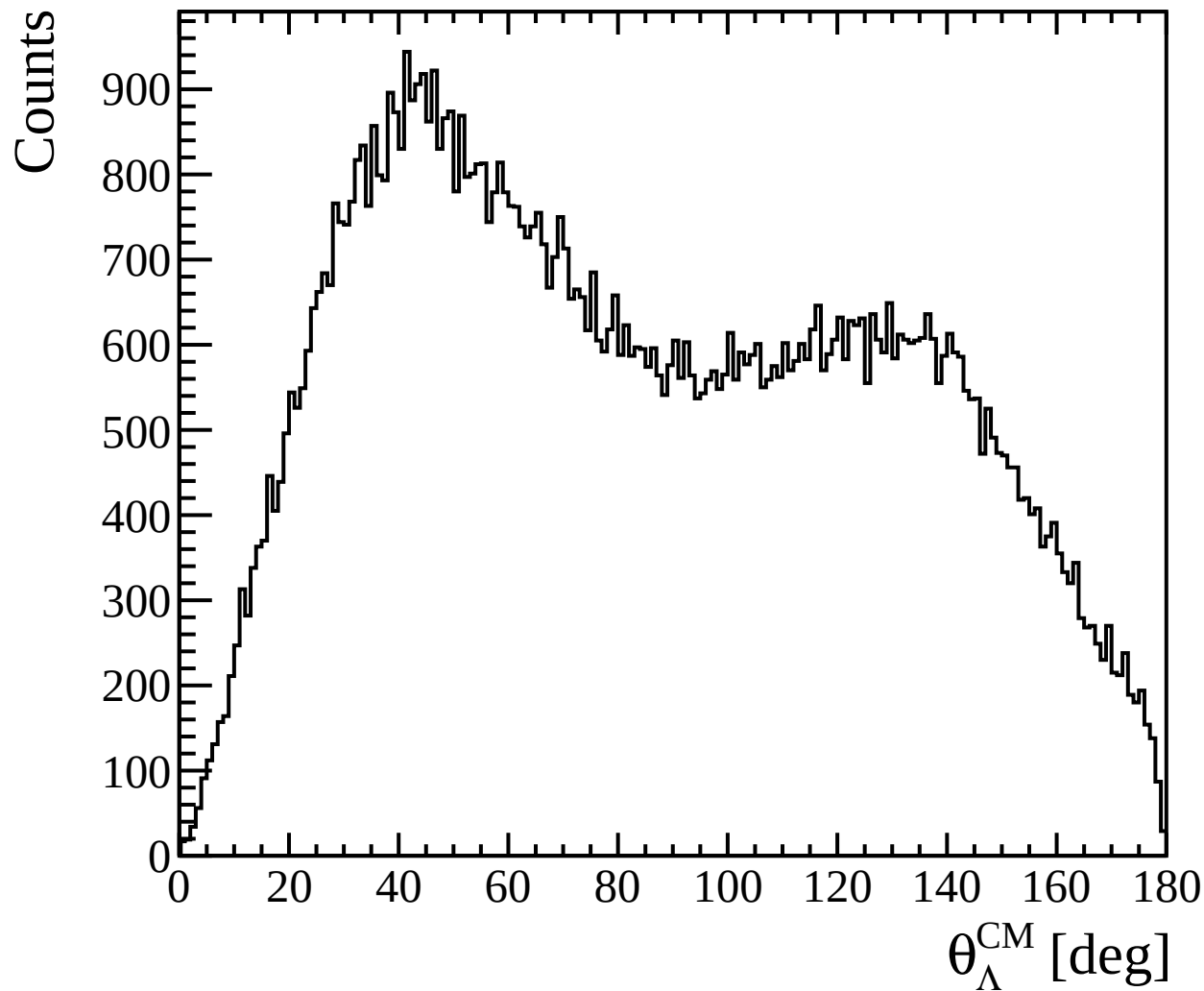


Reconstructed  $\Lambda$  production vertex (no LH2/Lambda-mass cut,  $M_x > 0.50$ )

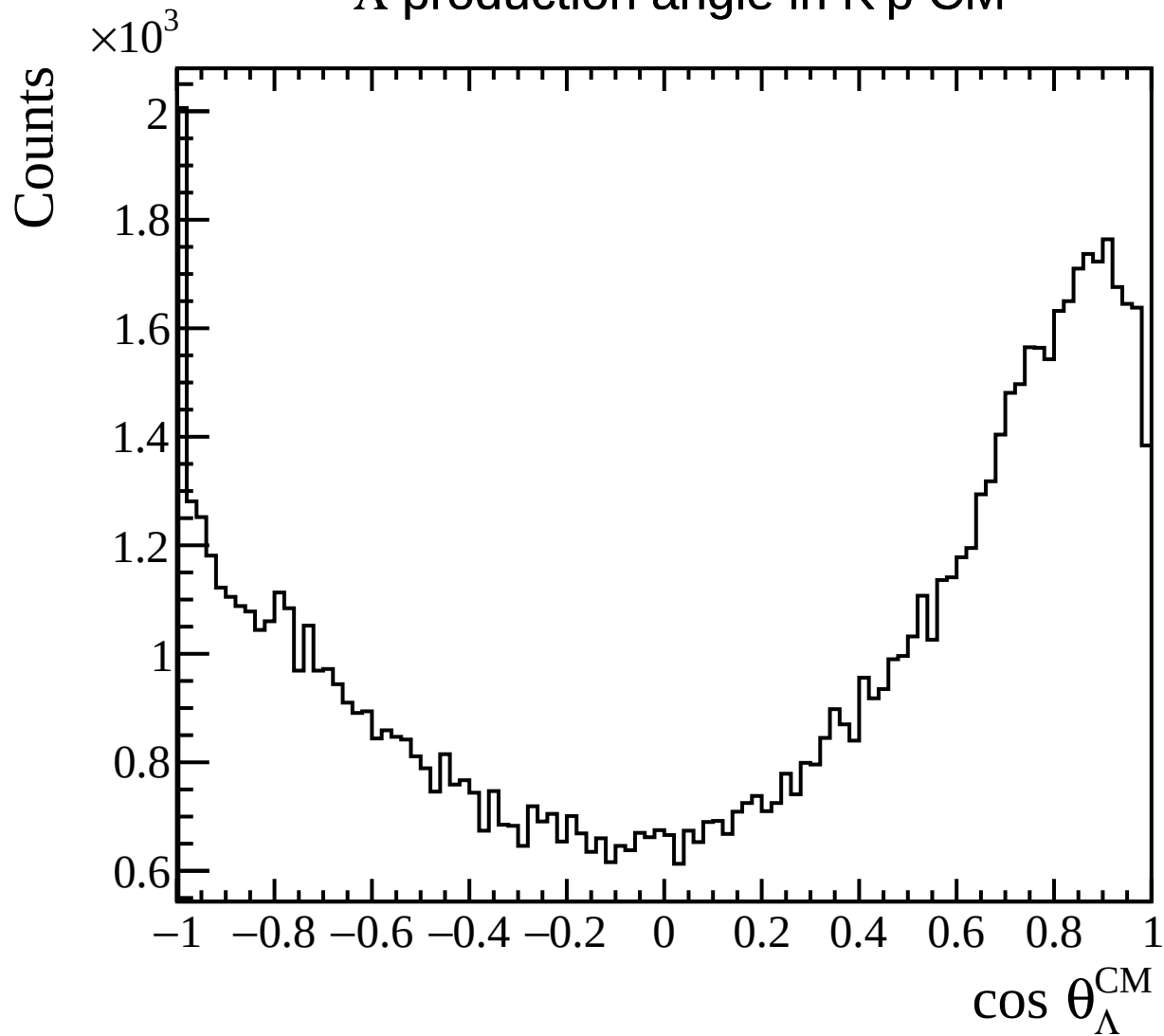




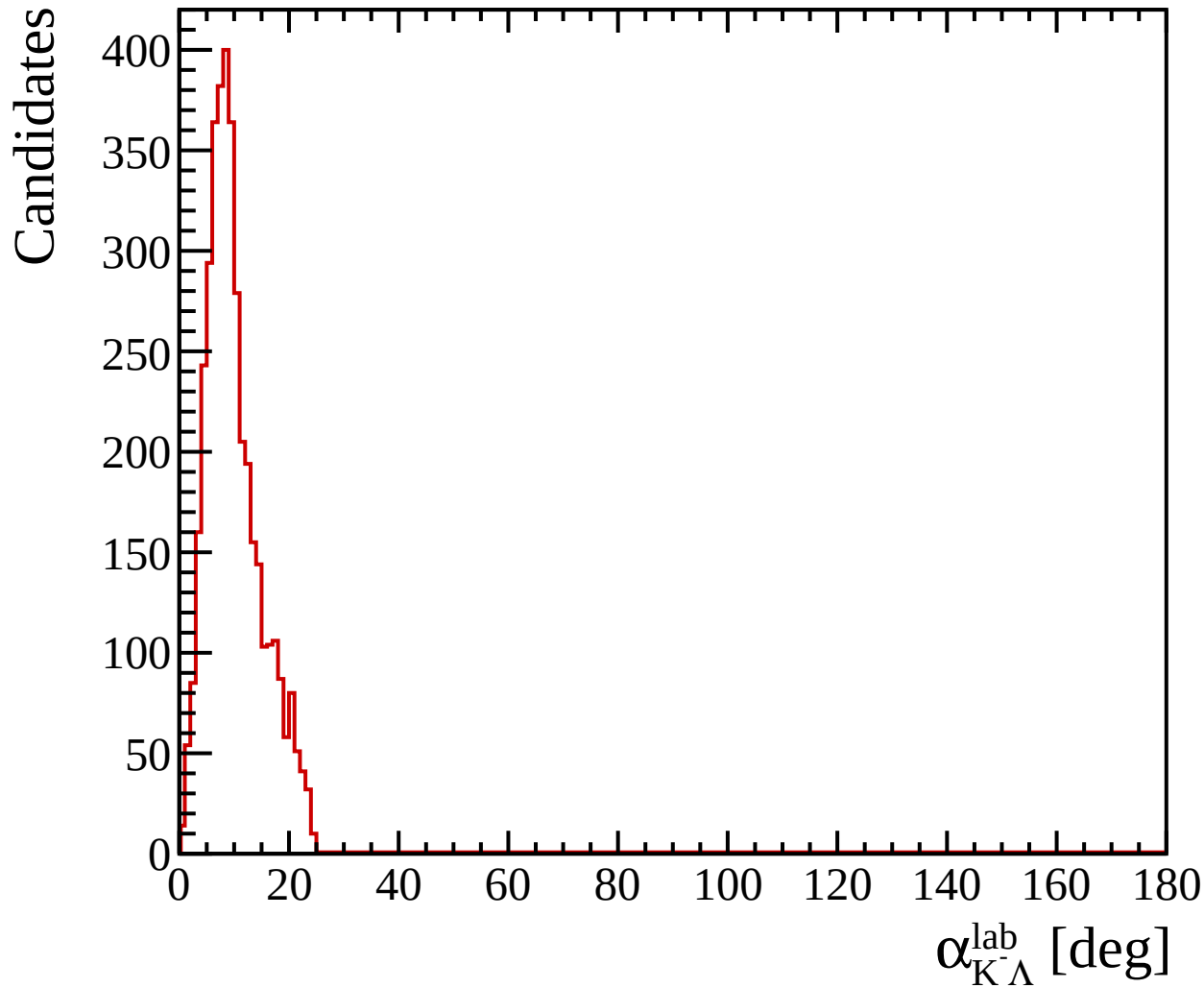
# $\Lambda$ production angle in $K^-p$ CM



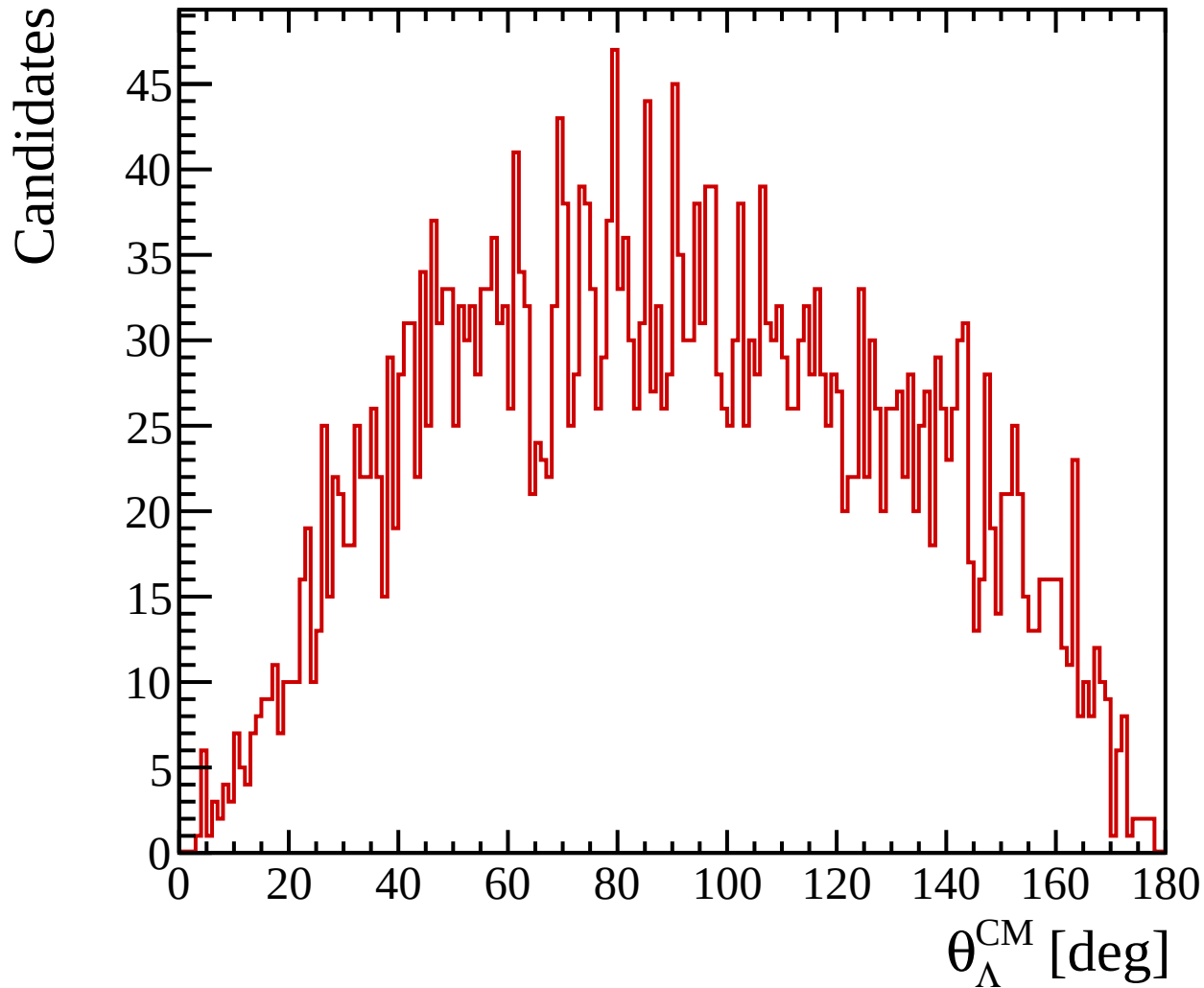
# $\Lambda$ production angle in $K^-p$ CM



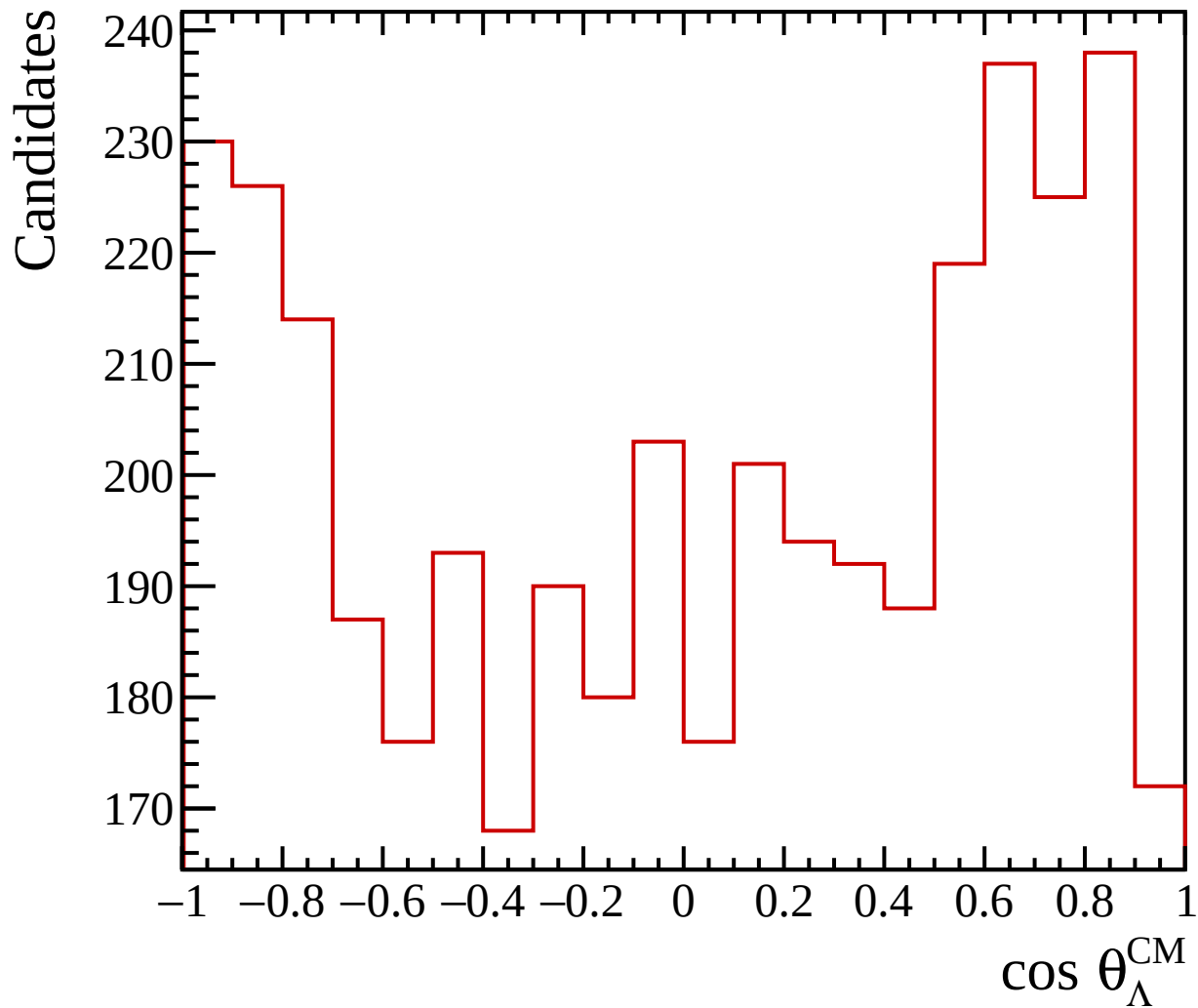
Selected  $\Lambda$  production angle ( $M_X > 0.50$ , LH2 inside)



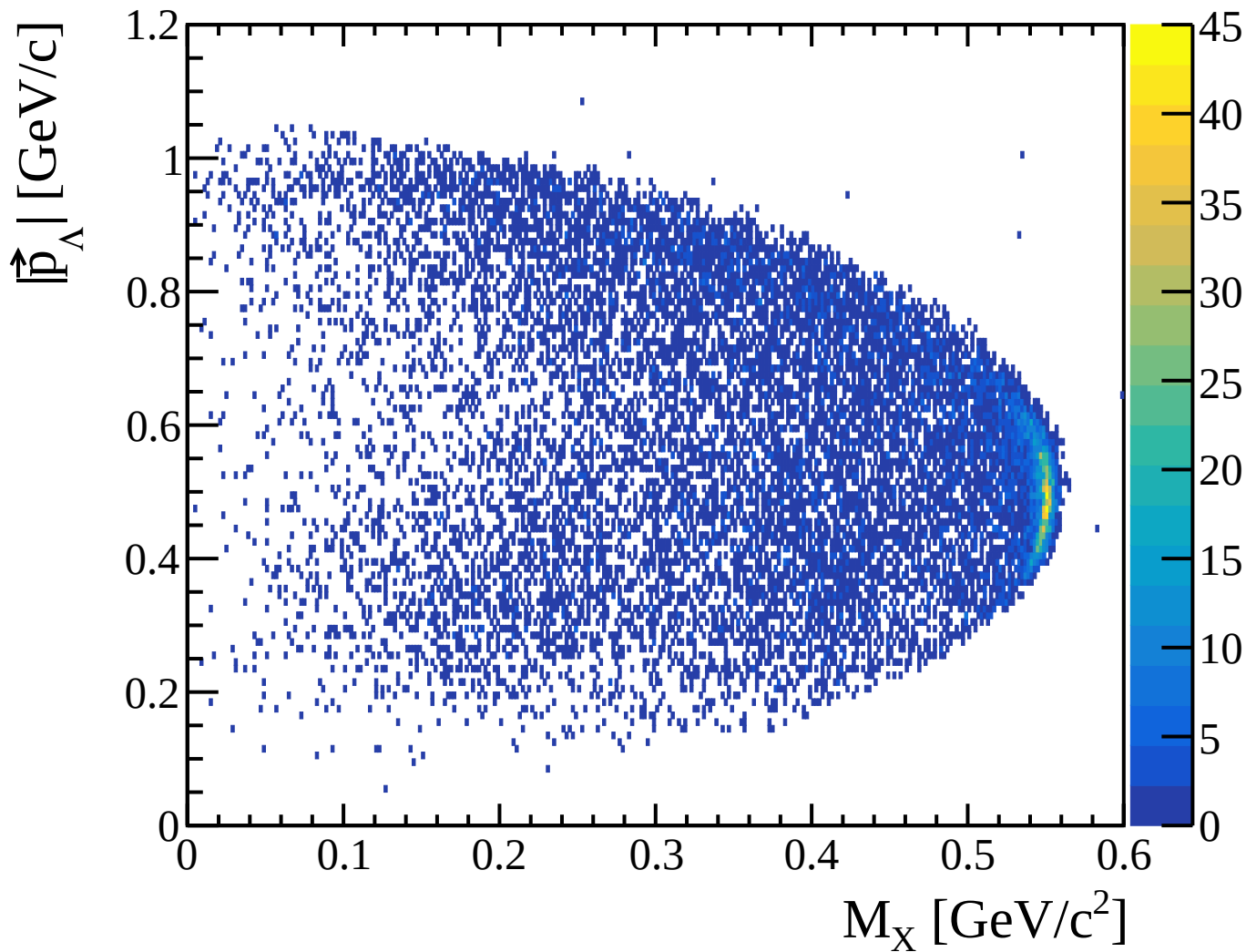
Selected  $\Lambda$  production angle ( $M_X > 0.50$ , LH2 inside)



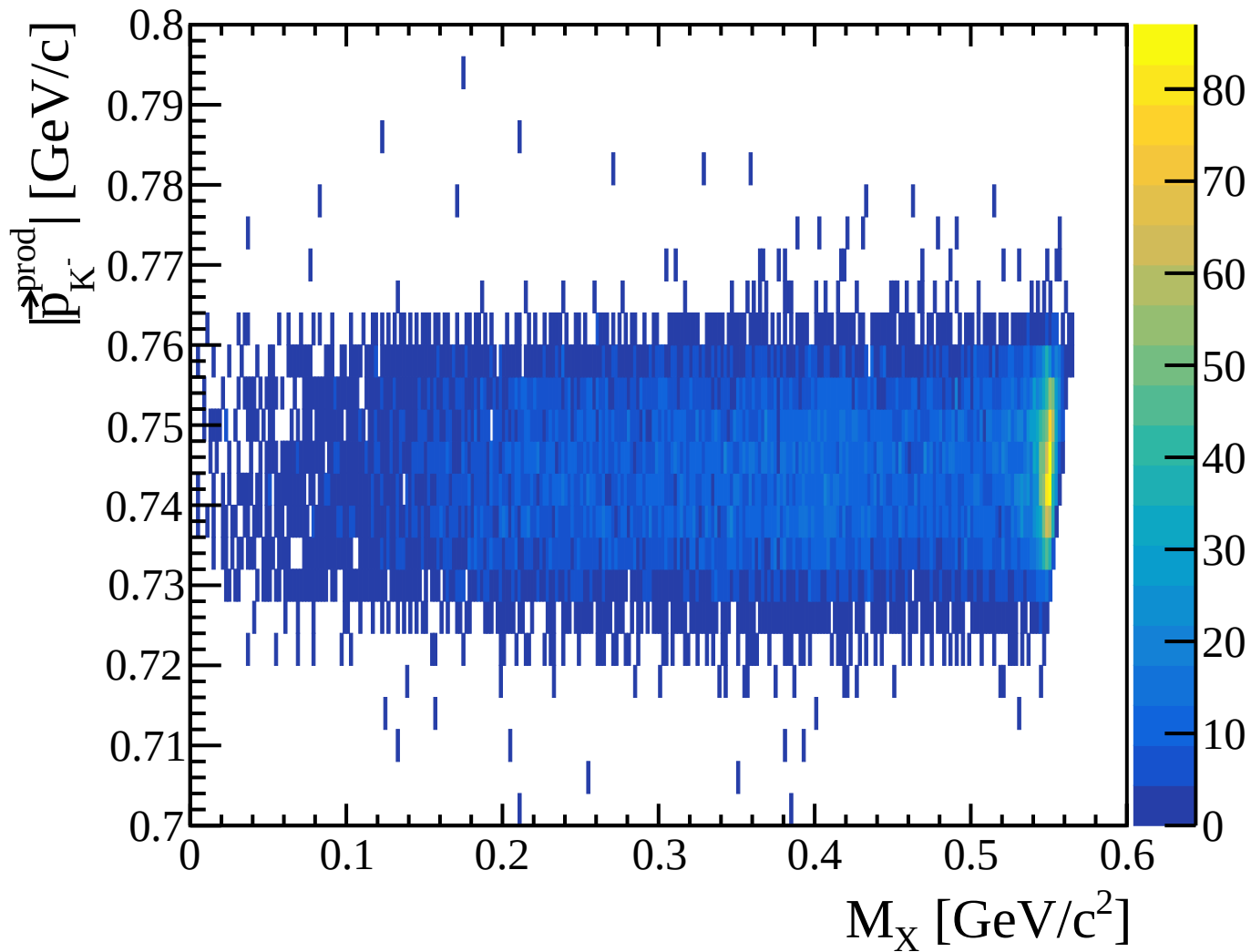
Selected  $\Lambda$  production angle ( $M_X > 0.50$ , LH2 inside)



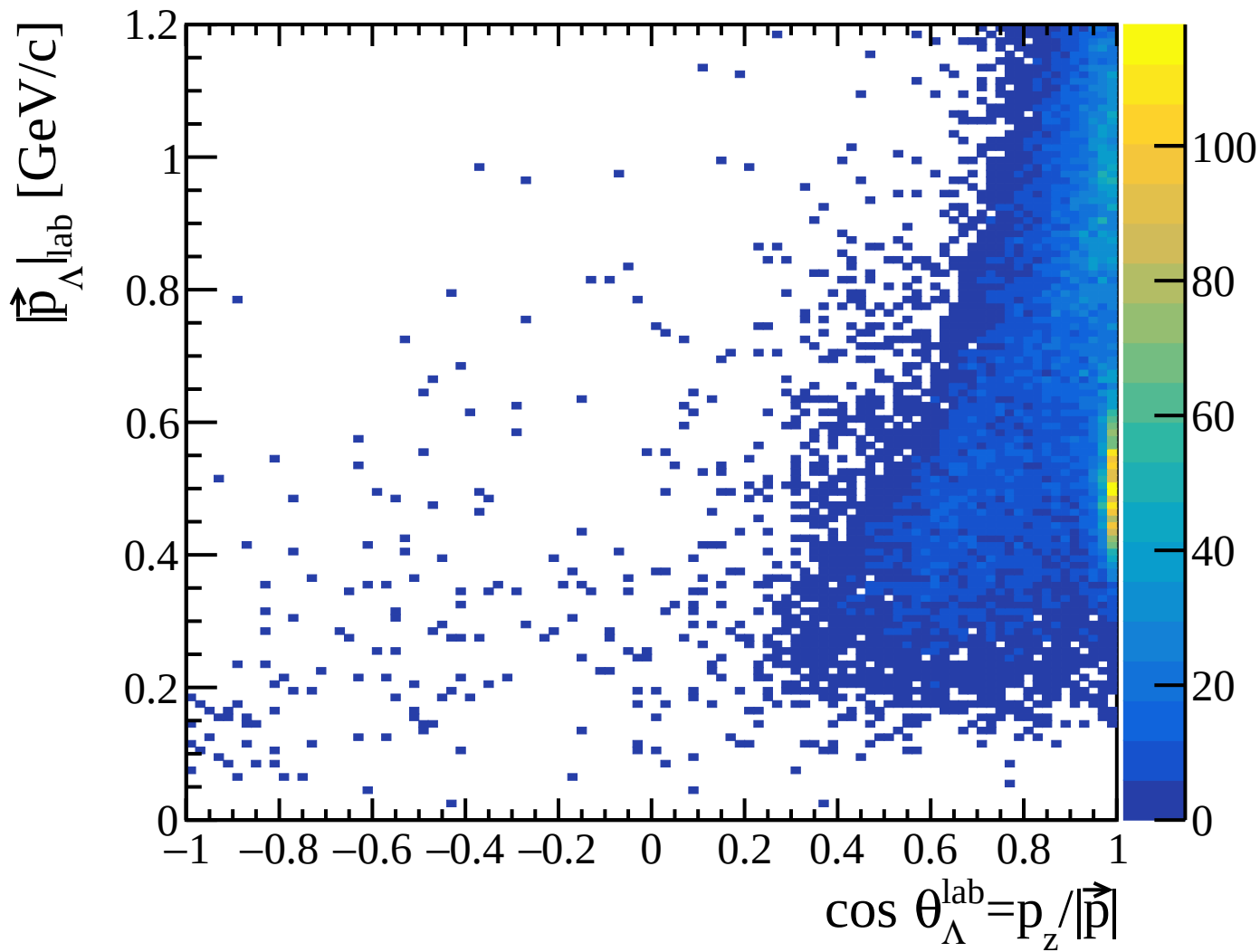
Missing mass vs reconstructed  $\Lambda$  momentum (LH2 inside, no  $M_X$  cut)



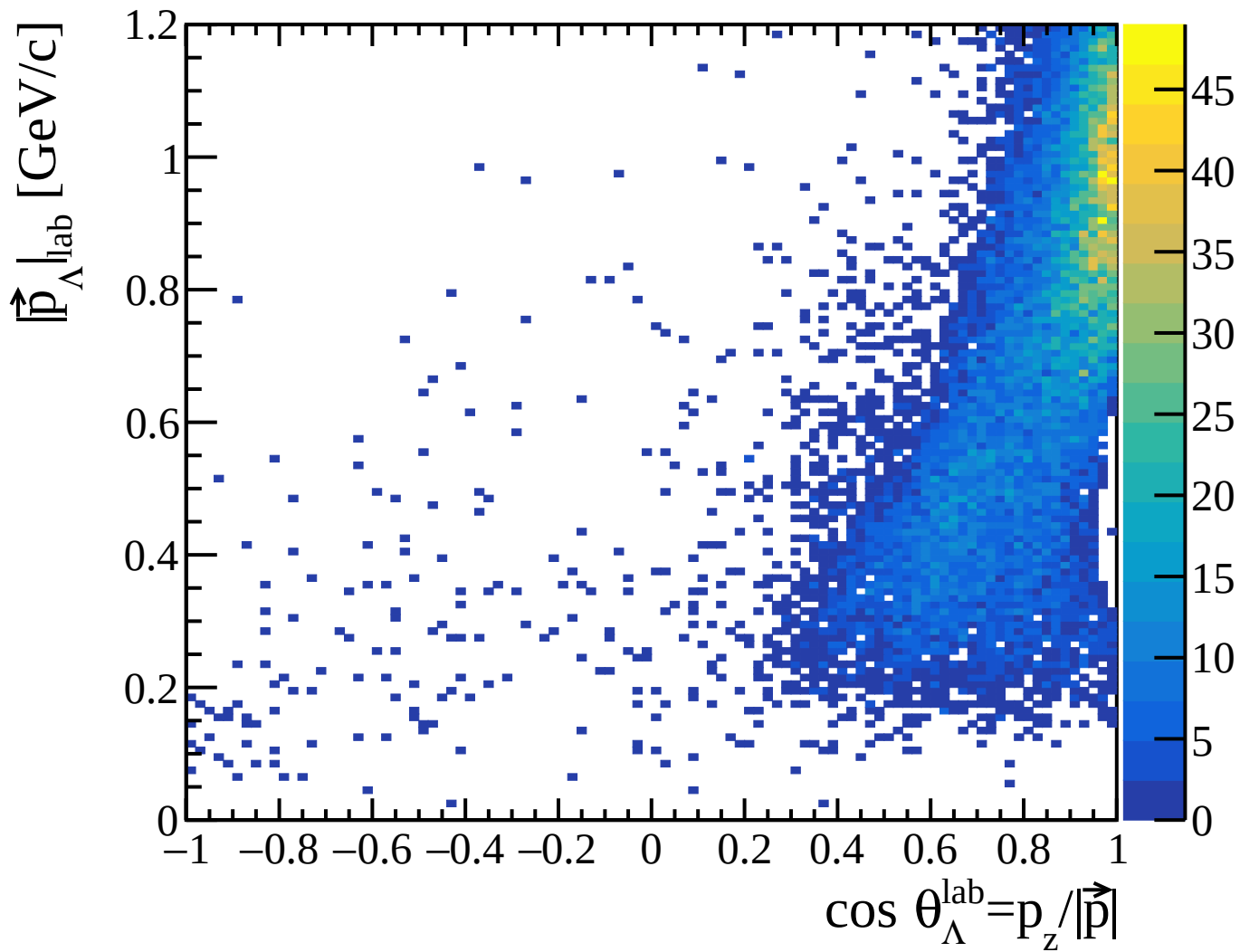
Missing mass vs production-vertex K18 momentum (LH2 inside, no  $M_X$  cut)



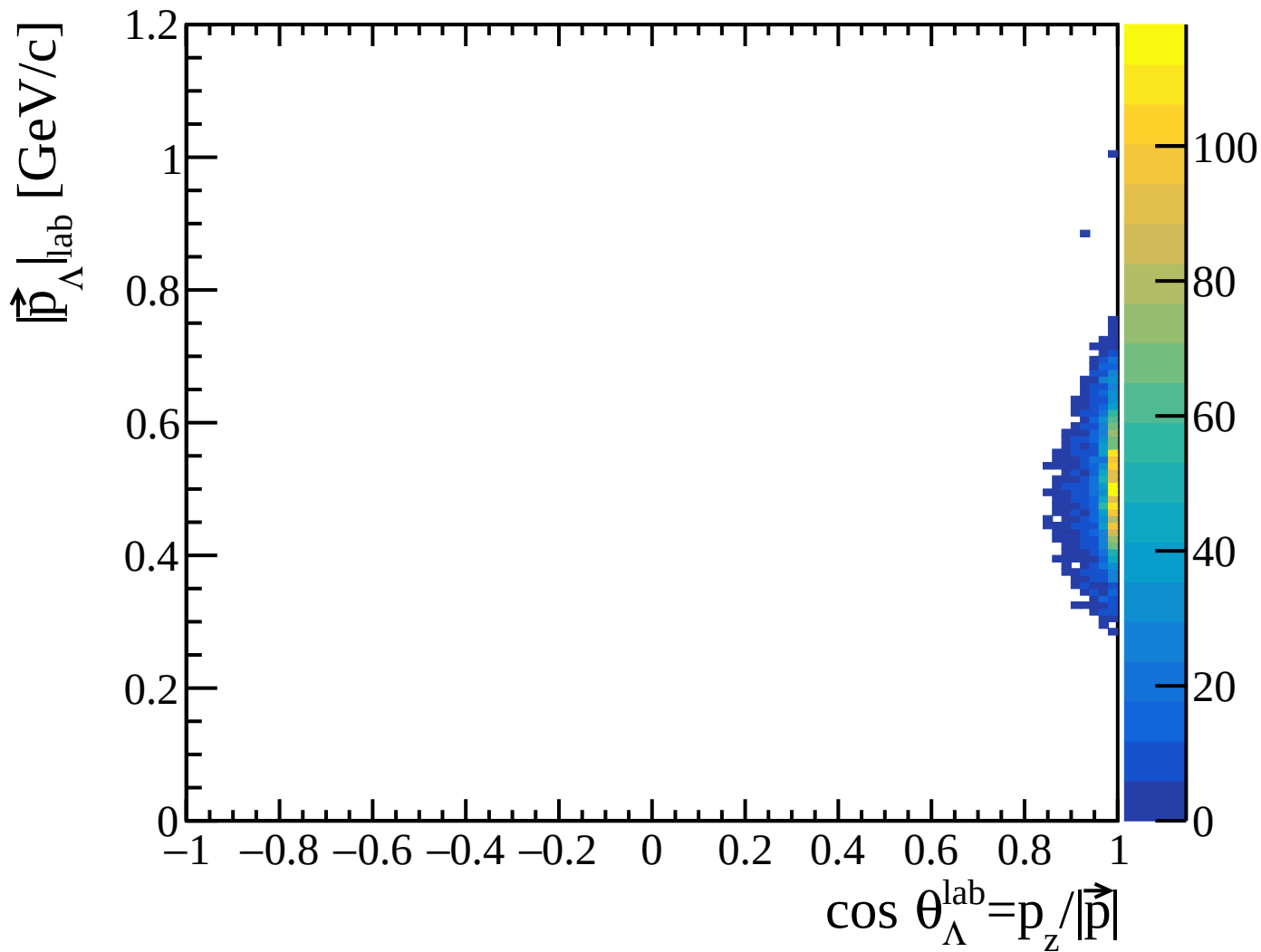
Selected  $\Lambda$ : lab  $\cos\theta$  vs lab momentum (LH2 inside, all  $M_X$ )



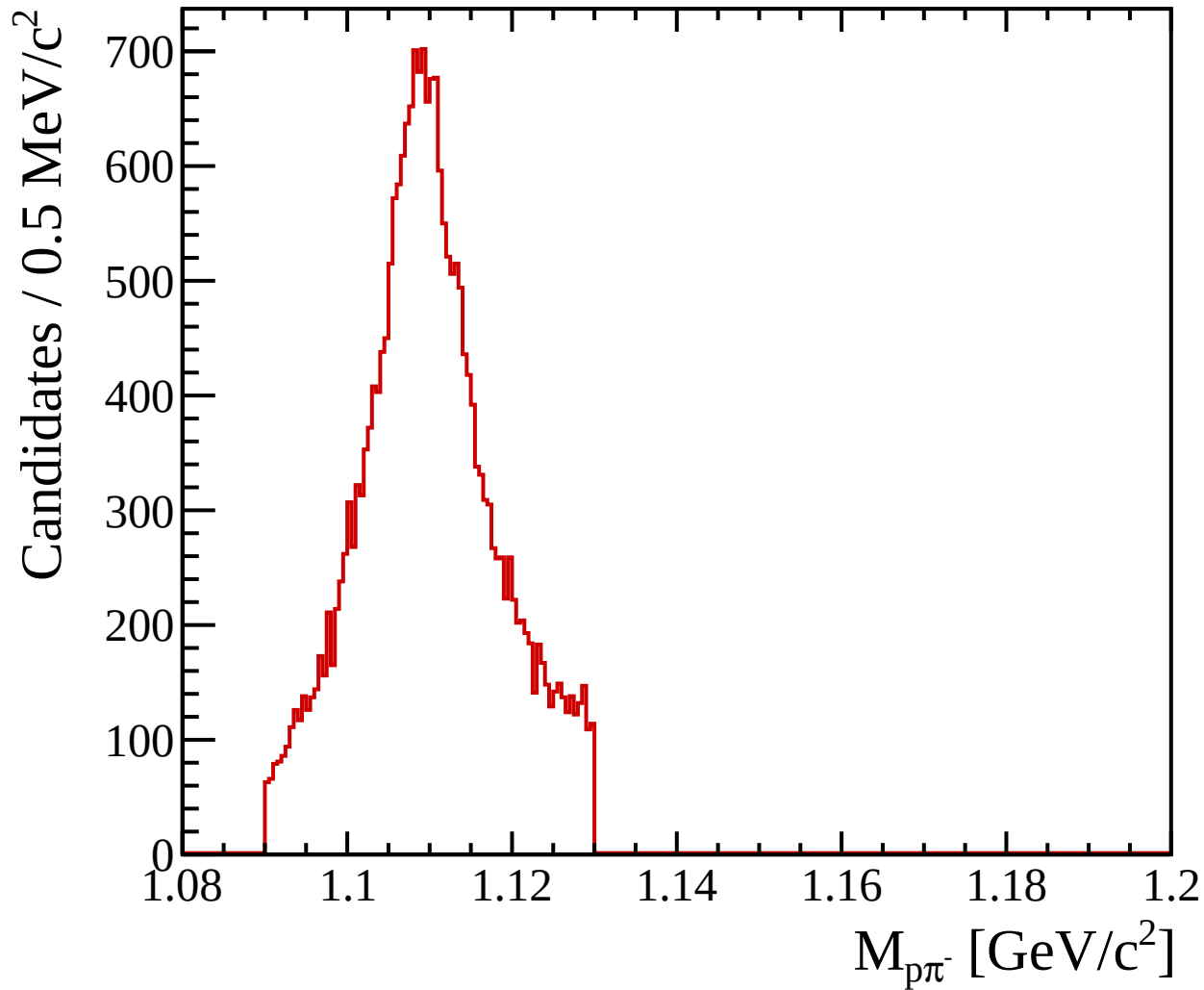
Selected  $\Lambda$ : lab  $\cos\theta$  vs lab momentum (LH2 inside,  $M_x \leq 0.50 \text{ GeV}/c^2$ )



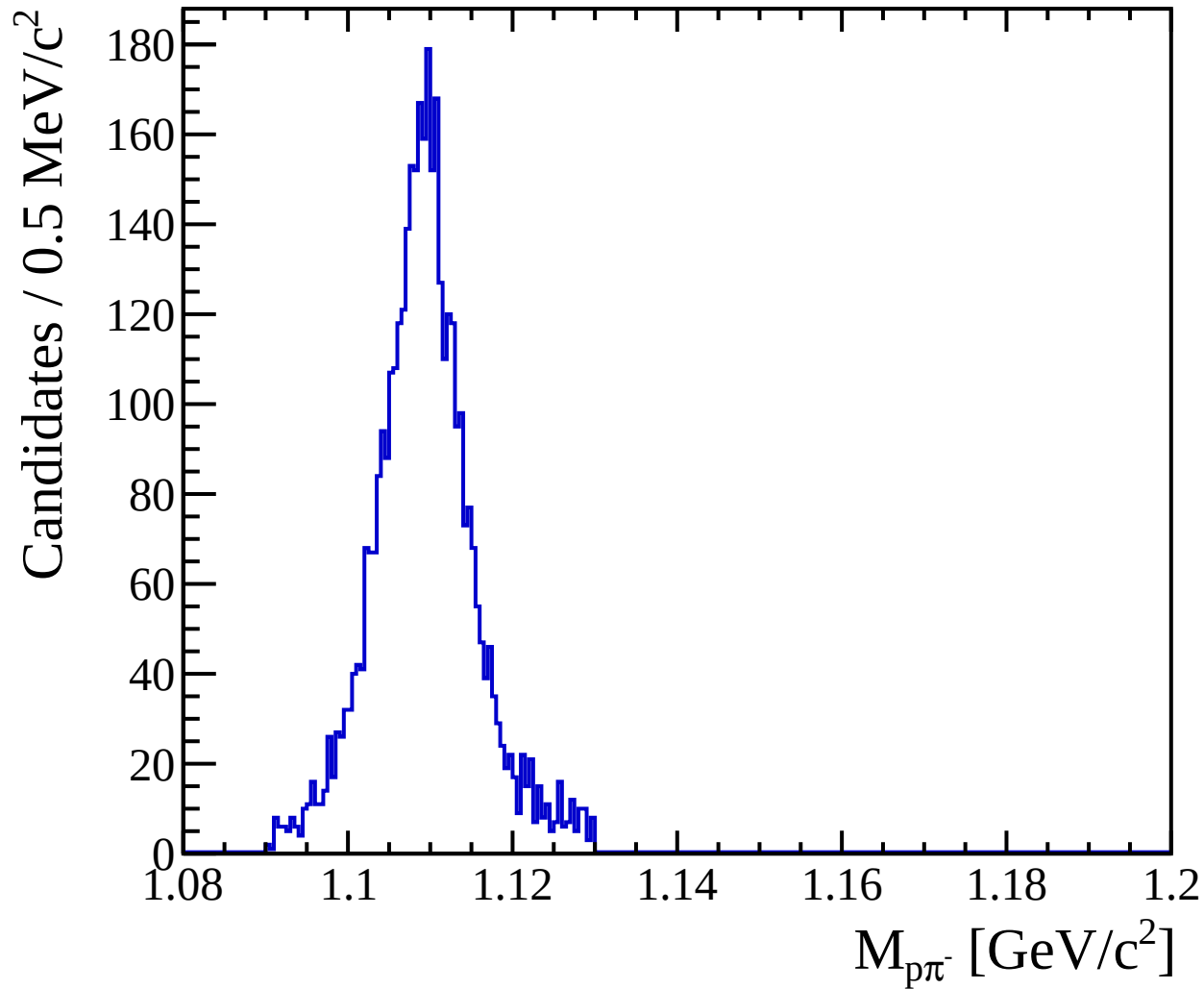
Selected  $\Lambda$ : lab  $\cos\theta$  vs lab momentum (LH2 inside,  $M_X > 0.50 \text{ GeV}/c^2$ )



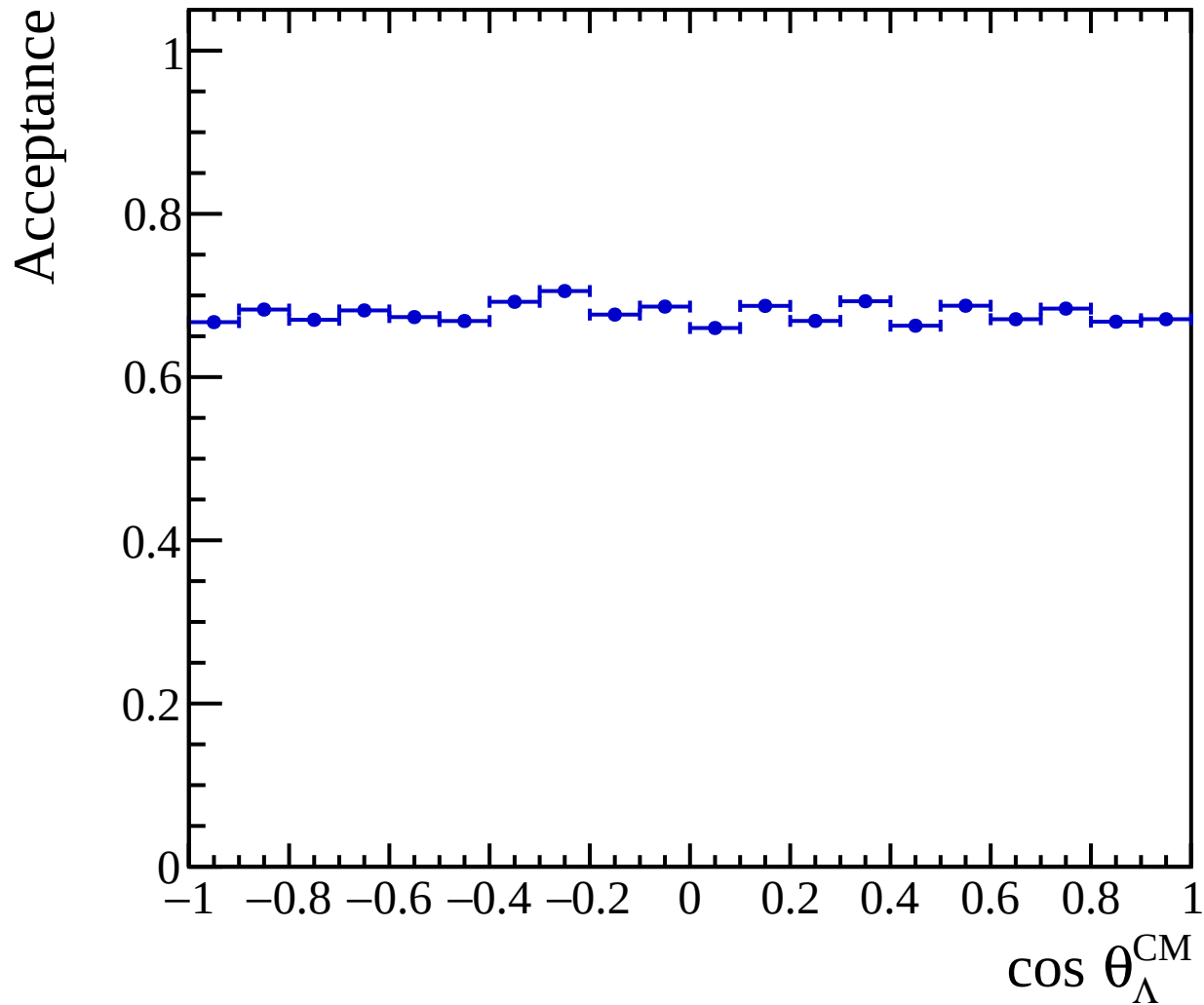
Selected  $\Lambda$  invariant mass (production vertex, LH2 inside)



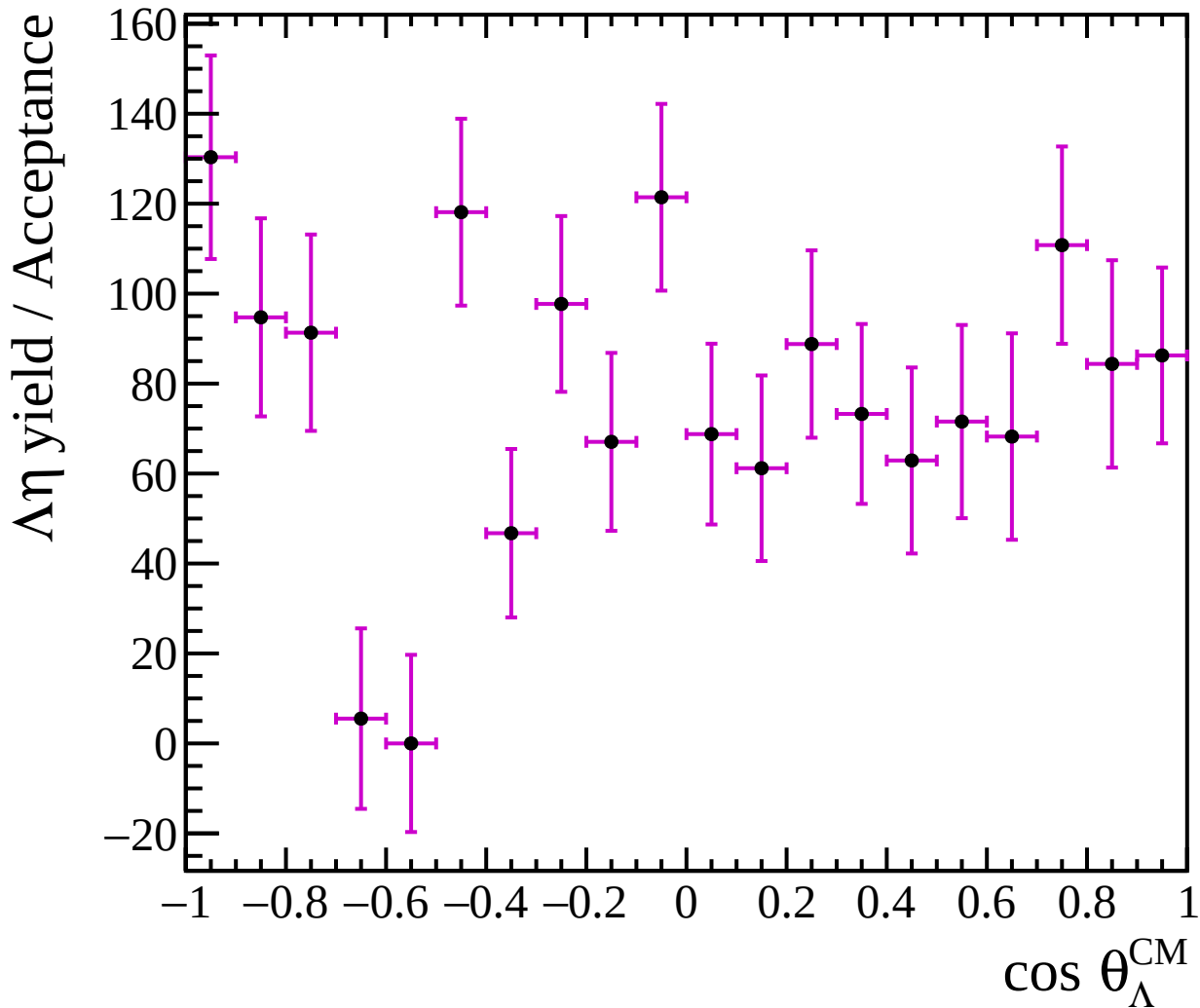
Selected  $\Lambda$  invariant mass (production vertex, LH2 inside,  $M_x > 0.50$ )

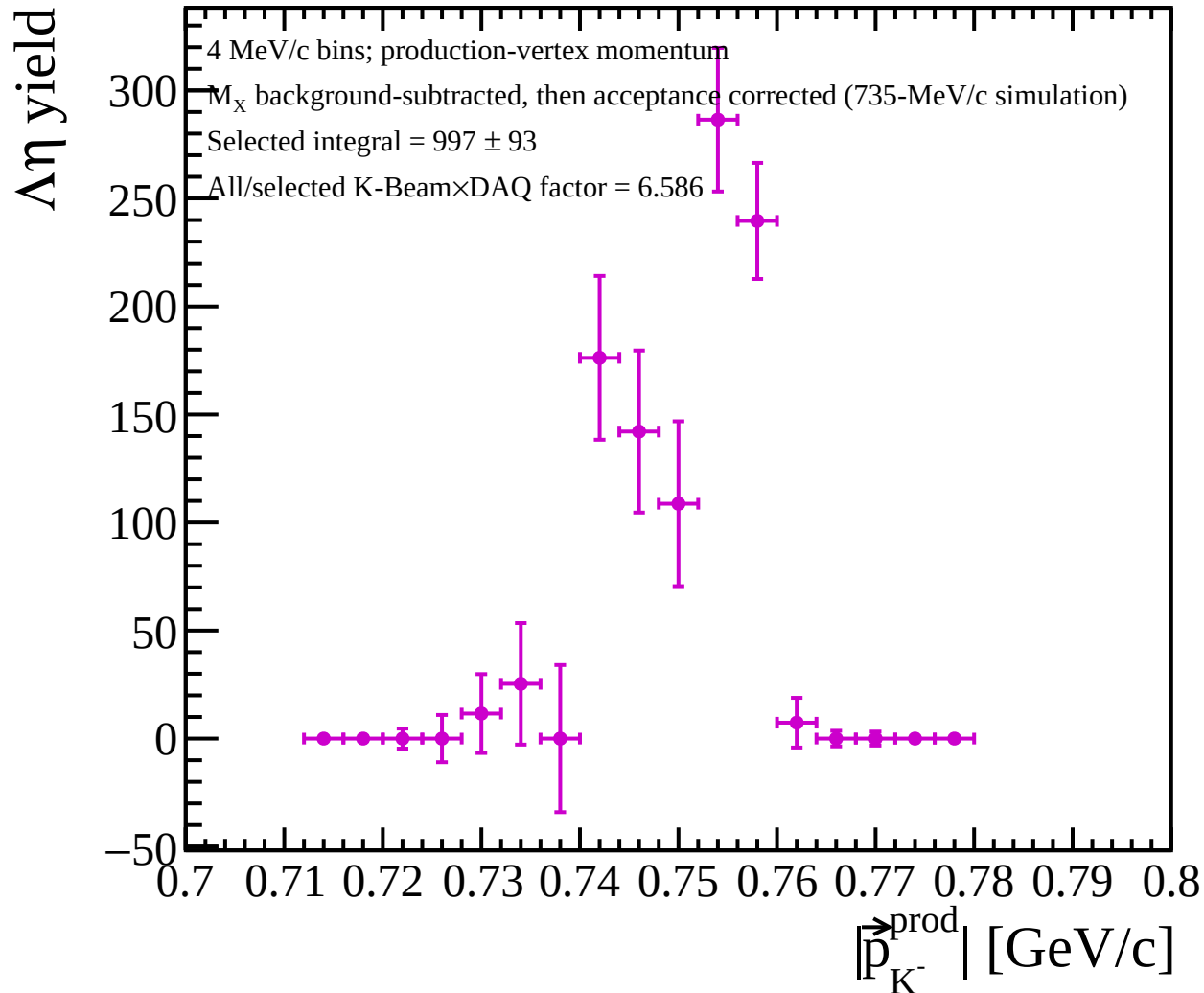


Trigger acceptance for generated  $\Lambda\eta$

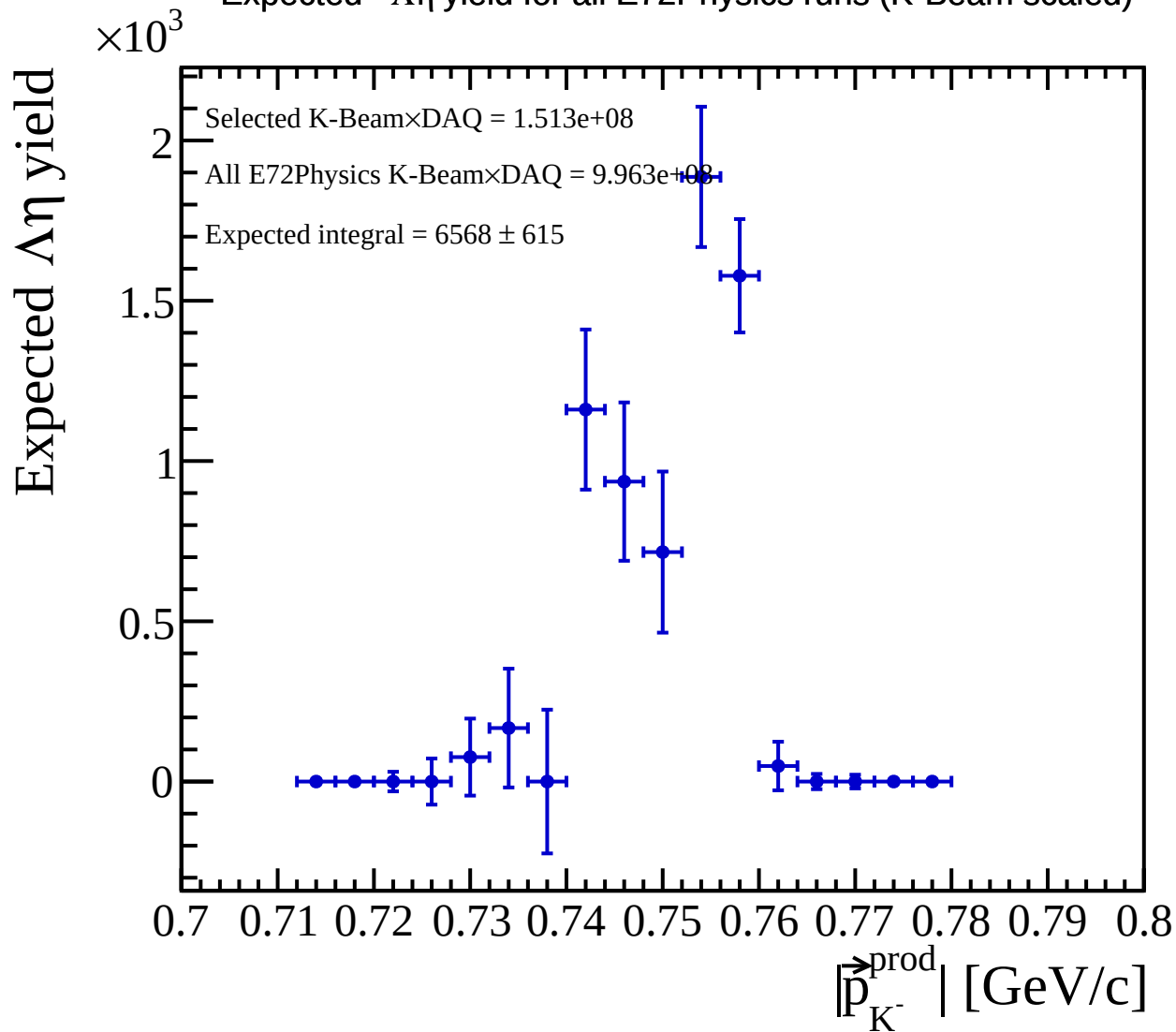


Background-subtracted  $\Lambda\eta$  production angle, trigger-acceptance corrected (LH2 inside,  $1.09 < M_{p\pi^-} < 1.13$ )

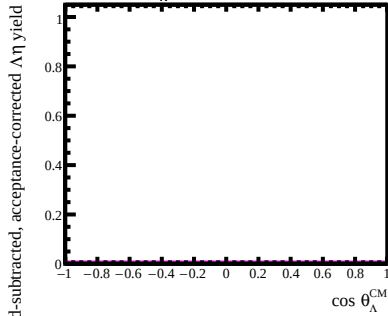




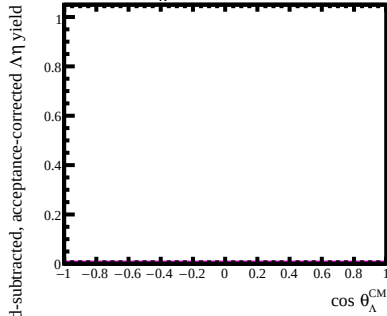
Expected  $\Lambda\eta$  yield for all E72Physics runs (K-Beam scaled)



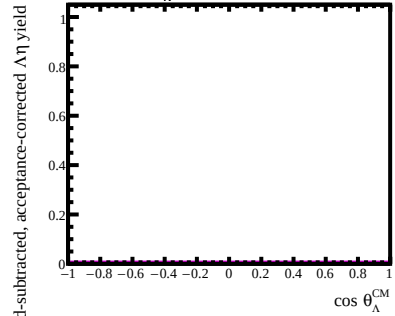
$p_K^{\text{prod}} = 700\text{--}704 \text{ MeV/c}$



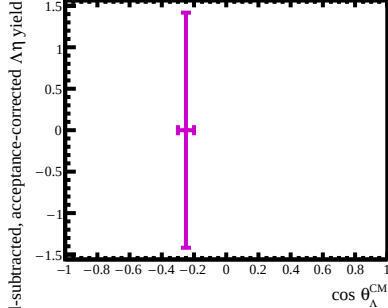
$p_K^{\text{prod}} = 704\text{--}708 \text{ MeV/c}$



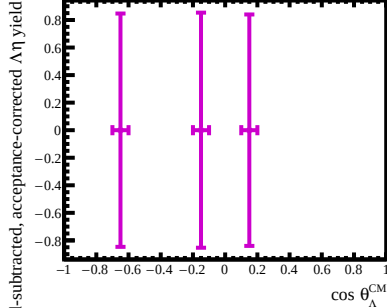
$p_K^{\text{prod}} = 708\text{--}712 \text{ MeV/c}$



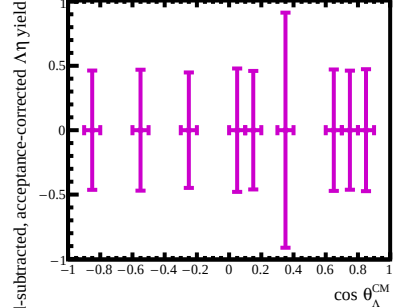
$p_K^{\text{prod}} = 712\text{--}716 \text{ MeV/c}$



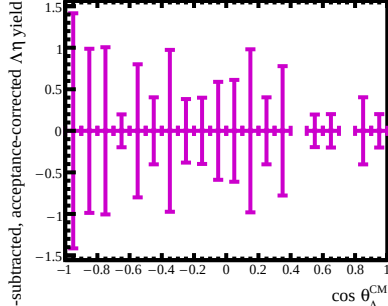
$p_K^{\text{prod}} = 716\text{--}720 \text{ MeV/c}$



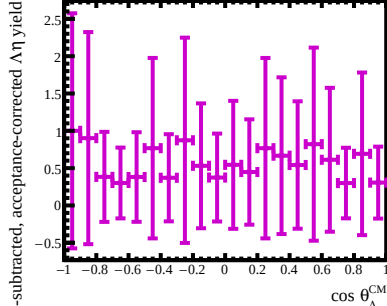
$p_K^{\text{prod}} = 720\text{--}724 \text{ MeV/c}$



$p_K^{\text{prod}} = 724\text{--}728 \text{ MeV/c}$



$p_K^{\text{prod}} = 728\text{--}732 \text{ MeV/c}$



$p_K^{\text{prod}} = 732\text{--}736 \text{ MeV/c}$

